

xtda+soc derivation

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CONTENTS

I. State interaction	3
II. Structure of excited sub-manifold	3
A. Basis of single excited manifold with $ S' = S + 1\rangle$	5
B. Basis of single excited manifold with $ S' = S\rangle$	5
C. Basis of single excited manifold with $ S' = S - 1\rangle$	6
D. Summary of excited states in state interaction	16
III. Derivation of SOC matrix element	17
A. Detailed derivation matrix elements of SOC operator	17
B. $ \Phi^{(S-1)(S-1)}\rangle$ interact with $ \Phi^{SS}\rangle$	17
C. $ \Phi^{SS}\rangle$ interact with $ \Phi^{(S+1)(S+1)}\rangle$	35
D. Summary of matrix elements	41
IV. Calculation of oscillator strength	46
A. $ S' = S - 1\rangle$ block	47
B. $ S' = S\rangle$ block	52
C. $ S' = S + 1\rangle$ block	54
D. Summary of matrix elements	54
Appendix: table of those excited states	57
References	59

I. STATE INTERACTION

After a TDA calculation, i.e. we solve a linear equation

$$\sum_{\nu} A_{\mu\nu} X_{\nu I} = \omega_I X_{\mu I}, \quad \text{for all } I \quad (1)$$

by diagonalizing coefficient matrix $(A_{\mu\nu})$. Then we construct a model space and take advantage of QDPT, we can consider the perturbation with the SOC operator

$$H_{IJ}^{\text{eff}} = \delta_{IJ} E_J^{(0)} + \langle \Psi_I | \hat{V} | \Psi_J \rangle, \quad (2)$$

here $|\Psi_I\rangle$ is excited state, which can be expanded in the basis of excited CSFs $\{|\Phi_J\rangle\}$

$$|\Psi_I\rangle = \sum_J \langle \Phi_J | \Psi_I \rangle |\Phi_J\rangle =: \sum_J X_{JI} |\Phi_J\rangle \quad (3)$$

the unknown thing in Eq. (2) is $\langle \Psi_I | \hat{V} | \Psi_J \rangle$, which is

$$\langle \Psi_I | \hat{V} | \Psi_J \rangle = \sum_{KL} X_{KI}^{\dagger} X_{LJ} \langle \Phi_K | \hat{V} | \Phi_L \rangle \quad (4)$$

since the coefficient matrix A is Hermitian, the CI coefficients X are real. Now we only need to determine the matrix elements $\langle \Phi_K | \hat{V} | \Phi_L \rangle$. The first task is to determine the explicit form of the states $|\Phi_L\rangle$ appearing on both sides.

II. STRUCTURE OF EXCITED SUB-MANIFOLD

Here the single excited manifold should be clarified, Specially, a triplet spin tensor operator $\hat{T}^{\dagger}(1)$ act on a tensor reference $|S\rangle\rangle$ reads

$$[\hat{T}^{\dagger}(1) \times |S\rangle\rangle]_{M'}^{S'} = \sum_{\mu=-1}^{+1} \hat{T}^{\dagger}(1, \mu) |S(M' - \mu)\rangle \times \langle 1\mu, S(M' - \mu) | S' M' \rangle \quad (5)$$

another important relationship is

$$[\hat{S}_-, \hat{O}^{\dagger}(I, \mu)] = \sqrt{I(I+1) - \mu(\mu-1)} \hat{O}^{\dagger}(I, \mu) \quad (6)$$

which means, we start from a high-spin reference $|SS\rangle$, we can obtain $|S(S-1)\rangle$ by

$$|S(S-1)\rangle = \frac{1}{\sqrt{S(S+1) - S(S-1)}} \hat{S}_- |SS\rangle = \frac{1}{\sqrt{2S}} \hat{S}_- |SS\rangle \quad (7)$$

and

$$\begin{aligned} |S(S-2)\rangle &= \frac{1}{\sqrt{S(S+1) - (S-1)(S-2)}} \hat{S}_- |S(S-1)\rangle \\ &= \frac{1}{2\sqrt{S(2S-1)}} (\hat{S}_-)^2 |SS\rangle \end{aligned} \quad (8)$$

The perturbation $\hat{V}$ can be chosen as

$$\hat{h}^{\text{SO}} = \sum_{pq} \left[h_{pq}^{-1} \hat{T}_{pq}^\dagger(1, 1) + h_{pq}^0 \hat{T}_{pq}^\dagger(1, 0) + h_{pq}^1 \hat{T}_{pq}^\dagger(1, -1) \right] \quad (9)$$

the singlet and triplet single excitation operators:

$$\hat{S}_{pq}^\dagger(0, 0) = \frac{1}{\sqrt{2}} (\hat{p}_\alpha^\dagger \hat{q}_\alpha + \hat{p}_\beta^\dagger \hat{q}_\beta) \quad (10)$$

$$\hat{T}_{pq}^\dagger(1, 1) = -\hat{p}_\alpha^\dagger \hat{q}_\beta \quad (11)$$

$$\hat{T}_{pq}^\dagger(1, 0) = \frac{1}{\sqrt{2}} (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) \quad (12)$$

$$\hat{T}_{pq}^\dagger(1, -1) = \hat{p}_\beta^\dagger \hat{q}_\alpha \quad (13)$$

these operators have the following property:

$$\left(\hat{T}_{pq}^\dagger(1, m) \right)^\dagger = (-1)^m \hat{T}_{qp}^\dagger(1, -m) \quad (14)$$

with the electronic integral

$$h_{pq}^{-1} = -iV_{pq}^x - V_{pq}^y \quad (15)$$

$$h_{pq}^0 = -i\sqrt{2}V_{pq}^z \quad (16)$$

$$h_{pq}^1 = iV_{pq}^x - V_{pq}^y \quad (17)$$

satisfy

$$(h_{pq}^{-1})^* = h_{pq}^1, \quad (h_{pq}^0)^* = -h_{pq}^0 \quad (18)$$

and $h_{pq}^m = -h_{qp}^m$, which is come from property of V_{pq}^l . The operator basis (after removing redundant and linearly dependence) for those excited states are

$$\hat{q}_{S'=S+1}^\dagger = \{\hat{T}_{\text{CV}}^\dagger(1)\} \quad (19)$$

$$\hat{q}_{S'=S}^\dagger = \{\hat{S}_{\text{CV}}^\dagger(0), \hat{S}_{\text{CO}}^\dagger(0), \hat{S}_{\text{OV}}^\dagger(0), \hat{T}_{\text{CV}}^\dagger(1)\} \quad (20)$$

$$\hat{q}_{S'=S-1}^\dagger = \{\hat{T}_{\text{CV}}^\dagger(1), \hat{T}_{\text{CO}}^\dagger(1), \hat{T}_{\text{OV}}^\dagger(1), \hat{T}_{\text{OO}}^\dagger(1)\} \quad (21)$$

A. Basis of single excited manifold with $|S' = S + 1\rangle$

$$a. \quad \hat{T}_{\text{CV}}^\dagger(1), |\Phi_{ai}^{(S+1)(S+1)}\rangle$$

$$\begin{aligned} [\hat{T}_{\text{CV}}^\dagger(1) \times |S\rangle]_{S+1}^{S+1} &= \hat{T}_{ai}^\dagger(1, 1)|SS\rangle \overbrace{\langle 11, SS|(S+1)(S+1)\rangle}^1 \\ &= \boxed{-\hat{a}_\alpha^\dagger \hat{i}_\beta |SS\rangle = -|\Phi_i^a\rangle} \end{aligned} \quad (22)$$

B. Basis of single excited manifold with $|S' = S\rangle$

$$a. \quad \hat{S}_{\text{CV}}^\dagger(0), |\Phi_{ai,1}^{SS}\rangle$$

$$[\hat{S}_{\text{CV}}^\dagger(0) \times |S\rangle]_S^S = \hat{S}_{ai}^\dagger(0, 0)|SS\rangle \overbrace{\langle 00, SS|SS\rangle}^1 = \boxed{\frac{1}{\sqrt{2}}(|\Phi_i^a\rangle + |\Phi_i^{\bar{a}}\rangle)} \quad (23)$$

$$b. \quad \hat{S}_{\text{CO}}^\dagger(0), |\Phi_{ui}^{SS}\rangle$$

$$[\hat{S}_{\text{CO}}^\dagger(0) \times |S\rangle]_S^S = \hat{S}_{ui}^\dagger(0, 0)|SS\rangle = \frac{1}{\sqrt{2}}(\hat{u}_\alpha^\dagger \hat{i}_\alpha + \hat{u}_\beta^\dagger \hat{i}_\beta)|\Phi_0\rangle = \frac{1}{\sqrt{2}}|\Phi_i^{\bar{u}}\rangle \quad (24)$$

then the orthonormalized excited state reads

$$\boxed{|\Phi_{ui,2}^{SS}\rangle = |\Phi_i^{\bar{u}}\rangle} \quad (25)$$

$$c. \quad \hat{S}_{\text{OV}}^\dagger(0), |\Phi_{au}^{SS}\rangle$$

$$[\hat{S}_{\text{OV}}^\dagger(0) \times |S\rangle]_S^S = \hat{S}_{au}^\dagger(0, 0)|SS\rangle = \frac{1}{\sqrt{2}}(\hat{a}_\alpha^\dagger \hat{u}_\alpha + \hat{a}_\beta^\dagger \hat{u}_\beta)|\Phi_0\rangle = \frac{1}{\sqrt{2}}|\Phi_u^a\rangle \quad (26)$$

then the orthonormalized excited state reads

$$\boxed{|\Phi_{au,2}^{SS}\rangle = |\Phi_u^a\rangle} \quad (27)$$

$$d. \quad \hat{T}_{\text{CV}}^\dagger(1), |\Phi_{ai,2}^{SS}\rangle$$

$$\begin{aligned}
[\hat{T}_{\text{CV}}^\dagger(1) \times |S\rangle]_S^S &= \hat{T}_{ai}^\dagger(1, 0) |SS\rangle \overbrace{\langle 10, SS | SS \rangle}^{-S/\sqrt{S(S+1)}} \\
&\quad + \hat{T}_{ai}^\dagger(1, 1) |S(S-1)\rangle \overbrace{\langle 11, S(S-1) | SS \rangle}^{1/\sqrt{S+1}} \\
&= \hat{T}_{ai}^\dagger(1, 0) |SS\rangle \frac{-S}{\sqrt{S(S+1)}} + \hat{T}_{ai}^\dagger(1, 1) \hat{S}_- |SS\rangle \frac{1}{\sqrt{2S(S+1)}} \\
&= -\frac{S}{\sqrt{2S(S+1)}} (|\Phi_i^a\rangle - |\Phi_i^{\bar{a}}\rangle) - \frac{1}{\sqrt{2S(S+1)}} \sum_x |\Phi_{ix}^{a\bar{x}}\rangle \quad (28)
\end{aligned}$$

$$= \left[-\sqrt{\frac{S}{2(S+1)}} (|\Phi_i^a\rangle - |\Phi_i^{\bar{a}}\rangle) + \frac{1}{S} \sum_x |\Phi_{ix}^{a\bar{x}}\rangle \right] \quad (29)$$

the norm of those configurations are

$$\langle \Phi_i^a | \Phi_j^b \rangle = \langle \Phi_i^{\bar{a}} | \Phi_j^{\bar{b}} \rangle = \delta_{ij} \delta_{ab} \quad (30)$$

$$\sum_{xy} \langle \Phi_{ix}^{a\bar{x}} | \Phi_{jy}^{b\bar{y}} \rangle = \sum_{xy} \delta_{xy} \delta_{ij} \delta_{ab} = (2S) \delta_{ij} \delta_{ab} \quad (31)$$

the norm of this state is

$$\langle \Phi_{ai,2}^{SS} | \Phi_{bj,2}^{SS} \rangle = \frac{S}{2(S+1)} \left(1 + 1 + \frac{2S}{S^2} \right) \delta_{ij} \delta_{ab} = \delta_{ij} \delta_{ab} \quad (32)$$

which means excited state Eq. (29) is already orthonormalized.

C. Basis of single excited manifold with $|S' = S - 1\rangle$

$$a. \quad \hat{T}_{\text{CV}}^\dagger(1), |\Phi_{ai}^{(S-1)(S-1)}\rangle$$

$$\begin{aligned}
[\hat{T}_{\text{CV}}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \hat{T}_{ai}^\dagger(1, 1) (\hat{S}_-)^2 |SS\rangle \frac{1}{2\sqrt{S(2S-1)}} \overbrace{\langle 11, S(S-2) | (S-1)(S-1) \rangle}^{1/\sqrt{S(2S+1)}} \\
&\quad + \hat{T}_{ai}^\dagger(1, 0) \hat{S}_- |SS\rangle \frac{1}{\sqrt{2S}} \overbrace{\langle 10, S(S-1) | (S-1)(S-1) \rangle}^{-\sqrt{(2S-1)/S(2S+1)}} \\
&\quad + \hat{T}_{ai}^\dagger(1, -1) |SS\rangle \overbrace{\langle 1-1, SS | (S-1)(S-1) \rangle}^{\sqrt{(2S-1)/(2S+1)}} \quad (33)
\end{aligned}$$

$$\begin{aligned}
&= \hat{T}_{ai}^\dagger(1,1)(\hat{S}_-)^2|SS\rangle \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \\
&\quad + \hat{T}_{ai}^\dagger(1,0)\hat{S}_-|SS\rangle \left(-\frac{1}{S}\sqrt{\frac{2S-1}{2(2S+1)}}\right) \\
&\quad + \hat{T}_{ai}^\dagger(1,-1)|SS\rangle \sqrt{\frac{2S-1}{2S+1}}
\end{aligned} \tag{34}$$

the first term is very simple, note that $x \neq y$ in this formula (if $x = y$, then $\hat{x}_\alpha \hat{x}_\beta^\dagger \hat{x}_\alpha |\Phi_0^{SS}\rangle = -\hat{x}_\beta^\dagger \hat{x}_\alpha \hat{x}_\alpha |\Phi_0^{SS}\rangle = 0$), then

$$\hat{T}_{ai}^\dagger(1,-1)(\hat{S}_-)^2|SS\rangle = -\hat{a}_\alpha^\dagger \hat{i}_\beta \sum_{xy} \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha |\Phi_0\rangle = -\sum_{xy} |\Phi_{i\bar{x}y}^{a\bar{x}y}\rangle \tag{35}$$

the second term is

$$\begin{aligned}
\hat{T}_{ai}^\dagger(1,0)\hat{S}_-|SS\rangle &= \frac{1}{\sqrt{2}}(\hat{a}_\alpha^\dagger \hat{i}_\alpha - \hat{a}_\beta^\dagger \hat{i}_\beta) \sum_x \hat{x}_\beta^\dagger \hat{x}_\alpha |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}} \sum_x (|\Phi_{ix}^{a\bar{x}}\rangle - |\Phi_{i\bar{x}}^{\bar{a}\bar{x}}\rangle)
\end{aligned} \tag{36}$$

and the third term

$$\hat{T}_{ai}^\dagger(1,-1)|SS\rangle = \hat{a}_\beta^\dagger \hat{i}_\alpha |\Phi_0\rangle = |\Phi_i^{\bar{a}}\rangle \tag{37}$$

finally, we obtain the expression of excited state with CV(1) excitation:

$$\begin{aligned}
|\Phi_{ai}^{(S-1)(S-1)}\rangle [\hat{T}_{ai}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \sqrt{\frac{2S-1}{2S+1}} |\Phi_i^{\bar{a}}\rangle - \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} \sum_x (|\Phi_{ix}^{a\bar{x}}\rangle - |\Phi_{i\bar{x}}^{\bar{a}\bar{x}}\rangle) \\
&\quad - \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \sum_{xy} |\Phi_{i\bar{x}y}^{a\bar{x}y}\rangle
\end{aligned} \tag{38}$$

the norm of the single configuration can be calculated very easy

$$\langle \Phi_i^{\bar{a}} | \Phi_j^{\bar{b}} \rangle = \langle \Phi_0 | \hat{i}_\alpha^\dagger \hat{a}_\beta \hat{b}_\beta^\dagger \hat{j}_\alpha | \Phi_0 \rangle = \delta_{ij} \delta_{ab} \tag{39}$$

$$\sum_{xy} \langle \Phi_{ix}^{a\bar{x}} | \Phi_{jy}^{b\bar{y}} \rangle = \sum_{xy} \delta_{xy} \delta_{ij} \delta_{ab} = 2S \delta_{ij} \delta_{ab} \tag{40}$$

$$\sum_{xy} \langle \Phi_{i\bar{x}}^{\bar{a}\bar{x}} | \Phi_{j\bar{y}}^{\bar{b}\bar{y}} \rangle = 2S \delta_{ij} \delta_{ab} \tag{41}$$

$$\sum_{xy} \langle \Phi_{ix}^{\bar{a}\bar{x}} | \Phi_{jy}^{b\bar{y}} \rangle = \sum_{xy} \langle \Phi_{ix}^{a\bar{x}} | \Phi_{jy}^{\bar{b}\bar{y}} \rangle = 0 \quad (42)$$

$$\begin{aligned} \sum_{wxyz} \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \Phi_{jzw}^{b\bar{z}\bar{w}} \rangle &= \sum_{wxyz} \langle \Phi_0 | \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{x}_\beta^\dagger \hat{a}_\alpha^\dagger \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{w}_\beta^\dagger \hat{w}_\alpha^\dagger \hat{z}_\alpha^\dagger \hat{j}_\beta | \Phi_0 \rangle \\ &= \delta_{ij} \delta_{ab} \sum_{wxyz} (\delta_{xz} \delta_{yw} + \delta_{xw} \delta_{yz}) = (2S)(2S-1) \delta_{ij} \delta_{ab} \end{aligned} \quad (43)$$

where notice that $x \neq y$ and $z \neq w$ in the formulae above, so the last equality gives $\sum_{x \neq y} 1 = (2S)(2S-1) \neq (2S)^2$. Then the norm of this excited state becomes

$$\begin{aligned} \langle \Phi_{ai}^{(S-1)(S-1)} | \Phi_{bj}^{(S-1)(S-1)} \rangle &= \left(\frac{2S-1}{2S+1} + \frac{4S}{4S^2} \frac{2S-1}{2S+1} + \frac{1}{4S^2} \frac{2 \cdot 2S(2S-1)}{(2S+1)(2S-1)} \right) \delta_{ij} \delta_{ab} \\ &= \left(\frac{S+1}{S} \frac{2S-1}{2S+1} + \frac{1}{S(2S+1)} \right) \delta_{ij} \delta_{ab} = \delta_{ij} \delta_{ab} \end{aligned} \quad (44)$$

which means the Eq. (38) is already the orthonormal basis in CV excited submanifold.

$$b. \quad \hat{T}_{\text{CO}}^\dagger(1), |\Phi_{ui}^{(S-1)(S-1)}\rangle$$

$$\begin{aligned} [\hat{T}_{\text{CO}}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \hat{T}_{ui}^\dagger(1, 1) (\hat{S}_-)^2 |SS\rangle \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \\ &\quad + \hat{T}_{ui}^\dagger(1, 0) \hat{S}_- |S_i S_i\rangle \left(-\frac{1}{S} \sqrt{\frac{2S-1}{2(2S+1)}} \right) \\ &\quad + \hat{T}_{ui}^\dagger(1, -1) |SS\rangle \sqrt{\frac{2S-1}{2S+1}} \end{aligned}$$

the first term can be simplified as

$$\begin{aligned} \hat{T}_{ui}^\dagger(1, 1) (\hat{S}_-)^2 |SS\rangle &= -\hat{u}_\alpha^\dagger \hat{i}_\beta (S_-)^2 |\Phi_0\rangle = -\hat{u}_\alpha^\dagger \hat{i}_\beta \sum_{pq} \hat{p}_\beta^\dagger \hat{p}_\alpha \hat{q}_\beta^\dagger \hat{q}_\alpha |\Phi_0\rangle \\ &= -\sum_{xy} \hat{u}_\alpha^\dagger \hat{i}_\beta \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha |\Phi_0\rangle = -\sum_{xy} \hat{i}_\beta \hat{x}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha |\Phi_0\rangle \\ &= -\sum_{xy} (\hat{i}_\beta \hat{x}_\beta^\dagger \delta_{xu} \hat{y}_\beta^\dagger \hat{y}_\alpha - \hat{i}_\beta \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{u}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha) |\Phi_0\rangle \\ &= -\sum_{xy} (\hat{i}_\beta \hat{x}_\beta^\dagger \delta_{xu} \hat{y}_\beta^\dagger \hat{y}_\alpha + \hat{i}_\beta \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{y}_\beta^\dagger \delta_{uy}) |\Phi_0\rangle = 2 \sum_x |\Phi_{ix}^{\bar{u}\bar{x}}\rangle \end{aligned} \quad (45)$$

where the target is moving $\hat{u}_\alpha^\dagger$ to the next of reference $|\Phi_0^{SS}\rangle$, and $\hat{u}_\alpha^\dagger|\Phi_0^{SS}\rangle = 0$. and the second term reads

$$\begin{aligned}
\hat{T}_{ui}^\dagger(1,0)\hat{S}_-|S_i S_i\rangle &= \frac{1}{\sqrt{2}}(\hat{u}_\alpha^\dagger \hat{i}_\alpha - \hat{u}_\beta^\dagger \hat{i}_\beta)\hat{S}_-|\Phi_0\rangle = \frac{1}{\sqrt{2}}(\hat{u}_\alpha^\dagger \hat{i}_\alpha - \hat{u}_\beta^\dagger \hat{i}_\beta) \sum_p \hat{p}_\beta^\dagger \hat{p}_\alpha |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}}(\hat{u}_\alpha^\dagger \hat{i}_\alpha - \hat{u}_\beta^\dagger \hat{i}_\beta) \sum_x \hat{x}_\beta^\dagger \hat{x}_\alpha |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}} \sum_x (\hat{u}_\alpha^\dagger \hat{i}_\alpha \hat{x}_\beta^\dagger \hat{x}_\alpha - \hat{u}_\beta^\dagger \hat{i}_\beta \hat{x}_\beta^\dagger \hat{x}_\alpha) |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}} \sum_x (\hat{i}_\alpha \hat{x}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{x}_\alpha - \hat{u}_\beta^\dagger \hat{i}_\beta \hat{x}_\beta^\dagger \hat{x}_\alpha) |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}} \sum_x (\delta_{xu} \hat{i}_\alpha \hat{x}_\beta^\dagger - \hat{i}_\alpha \hat{x}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{x}_\alpha - \hat{u}_\beta^\dagger \hat{i}_\beta \hat{x}_\beta^\dagger \hat{x}_\alpha) |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}} (-|\Phi_i^{\bar{u}}\rangle - \sum_x |\Phi_{ix}^{\bar{u}\bar{x}}\rangle)
\end{aligned} \tag{46}$$

finally, we obtain

$$\begin{aligned}
[\hat{T}_{ui}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \sum_x |\Phi_{ix}^{\bar{u}\bar{x}}\rangle \\
&\quad + \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} (|\Phi_i^{\bar{u}}\rangle + \sum_x |\Phi_{ix}^{\bar{u}\bar{x}}\rangle) + \sqrt{\frac{2S-1}{2S+1}} |\Phi_i^{\bar{u}}\rangle \\
&= \frac{1}{S} \sqrt{\frac{2S+1}{2S-1}} \sum_x |\Phi_{ix}^{\bar{u}\bar{x}}\rangle + \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} |\Phi_i^{\bar{u}}\rangle + \sqrt{\frac{2S-1}{2S+1}} |\Phi_i^{\bar{u}}\rangle \\
&= \frac{1}{2S} \sqrt{(2S+1)(2S-1)} |\Phi_i^{\bar{u}}\rangle + \frac{1}{2S} \sqrt{\frac{2S+1}{2S-1}} \sum_x |\Phi_{ix}^{\bar{u}\bar{x}}\rangle
\end{aligned} \tag{47}$$

$$\tag{48}$$

the norm of $|\Phi_i^{\bar{u}}\rangle$ and $|\Phi_{ix}^{\bar{u}\bar{x}}\rangle$ reads

$$\langle \Phi_i^{\bar{u}} | \Phi_j^{\bar{v}} \rangle = \delta_{ij} \delta_{uv} \tag{49}$$

$$\sum_{\substack{x \neq u \\ y \neq v}} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \Phi_{jy}^{\bar{v}\bar{y}} \rangle = \sum_{\substack{x \neq u \\ y \neq v}} \delta_{xy} \delta_{ij} \delta_{uv} = (2S-1) \delta_{ij} \delta_{uv} \tag{50}$$

then the norm of CO excited state becomes

$$\langle \Phi_{ai}^{(S-1)(S-1)} | \Phi_{bj}^{(S-1)(S-1)} \rangle = \left(\frac{(2S+1)(2S-1)}{(2S)^2} + \frac{2S-1}{(2S)^2} \frac{2S+1}{2S-1} \right) \delta_{ij} \delta_{uv}$$

$$= \frac{2S+1}{2S} \delta_{ij} \delta_{uv} \quad (51)$$

then the orthonormal excited state becomes

$$|\Phi_{ui}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}} |\Phi_i^{\bar{u}}\rangle + \frac{1}{\sqrt{2S(2S-1)}} \sum_{x \neq u} |\Phi_{ix}^{u\bar{x}}\rangle \quad (52)$$

c. $\hat{T}_{\text{OV}}^\dagger(1), |\Phi_{au}^{(S-1)(S-1)}\rangle$

$$\begin{aligned} [\hat{T}_{\text{OV}}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \hat{T}_{au}^\dagger(1, 1) (\hat{S}_-)^2 |SS\rangle \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \\ &\quad + \hat{T}_{au}^\dagger(1, 0) \hat{S}_- |S_i S_i\rangle \left(-\frac{1}{S} \sqrt{\frac{2S-1}{2(2S+1)}}\right) \\ &\quad + \hat{T}_{au}^\dagger(1, -1) |SS\rangle \sqrt{\frac{2S-1}{2S+1}} \end{aligned}$$

the first term is

$$\begin{aligned} \hat{T}_{au}^\dagger(1, 1) (\hat{S}_-)^2 |SS\rangle &= -\hat{a}_\alpha^\dagger \hat{u}_\beta \sum_{xy} \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha |\Phi_0\rangle \\ &= -\sum_{xy} (\hat{a}_\alpha^\dagger \delta_{xu} \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha + \hat{a}_\alpha^\dagger \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{u}_\beta \hat{y}_\beta^\dagger \hat{y}_\alpha) |\Phi_0\rangle \\ &= -\sum_{xy} (\hat{a}_\alpha^\dagger \delta_{xu} \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha + \hat{a}_\alpha^\dagger \hat{x}_\beta^\dagger \hat{x}_\alpha \delta_{yu} \hat{y}_\alpha) |\Phi_0\rangle = -2 \sum_x |\Phi_{ux}^{a\bar{x}}\rangle \end{aligned} \quad (53)$$

the second term is

$$\begin{aligned} \hat{T}_{au}^\dagger(1, 0) \hat{S}_- |SS\rangle &= \frac{1}{\sqrt{2}} (\hat{a}_\alpha^\dagger \hat{u}_\alpha - \hat{a}_\beta^\dagger \hat{u}_\beta) \sum_x \hat{x}_\beta^\dagger \hat{x}_\alpha |\Phi_0\rangle \\ &= \frac{1}{\sqrt{2}} (-|\Phi_u^{\bar{a}}\rangle + \sum_x |\Phi_{ux}^{a\bar{x}}\rangle) \end{aligned} \quad (54)$$

the third term is

$$\hat{T}_{au}^\dagger(1, -1) |SS\rangle = \hat{a}_\beta^\dagger \hat{u}_\alpha |\Phi_0\rangle = |\Phi_u^{\bar{a}}\rangle \quad (55)$$

finally,

$$[\hat{T}_{au}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} = -\sum_x \frac{2}{2S(2S+1)(2S-1)} |\Phi_{ux}^{a\bar{x}}\rangle + \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} |\Phi_u^{\bar{a}}\rangle$$

$$\begin{aligned}
& -\frac{1}{2S}\sqrt{\frac{2S-1}{2S+1}}\sum_x|\Phi_{ux}^{a\bar{x}}\rangle + \sqrt{\frac{2S-1}{2S+1}}|\Phi_u^{\bar{a}}\rangle \\
& = \frac{1}{2S}\sqrt{(2S+1)(2S-1)}|\Phi_u^{\bar{a}}\rangle - \frac{1}{2S}\sqrt{\frac{2S+1}{2S-1}}\sum_x|\Phi_{ux}^{a\bar{x}}\rangle \quad (56)
\end{aligned}$$

then the norm of $|\Phi_u^{\bar{a}}\rangle$ and $|\Phi_{ux}^{a\bar{x}}\rangle$ are

$$\langle\Phi_u^{\bar{a}}|\Phi_v^{\bar{b}}\rangle = \delta_{uv}\delta_{ab} \quad (57)$$

$$\sum_{\substack{x\neq u \\ y\neq v}}\langle\Phi_{ux}^{a\bar{x}}|\Phi_{vy}^{b\bar{y}}\rangle = \sum_{\substack{x\neq u \\ y\neq v}}\delta_{xy}\delta_{uv}\delta_{ab} = (2S-1)\delta_{uv}\delta_{ab} \quad (58)$$

then the orthonormal OV excited state becomes

$$|\Phi_{au}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}}|\Phi_u^{\bar{a}}\rangle - \frac{1}{\sqrt{2S(2S-1)}}\sum_{x\neq u}|\Phi_{ux}^{a\bar{x}}\rangle \quad (59)$$

$$d. \quad \hat{T}_{O_1O_2}^\dagger(1), |\Phi_{uv}^{(S-1)(S-1)}\rangle$$

$$\begin{aligned}
[\hat{T}_{O_1O_2}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \hat{T}_{uv}^\dagger(1,1)(\hat{S}_-)^2|SS\rangle \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \\
&+ \hat{T}_{uv}^\dagger(1,0)\hat{S}_-|S_iS_i\rangle \left(-\frac{1}{S}\sqrt{\frac{2S-1}{2(2S+1)}}\right) \\
&+ \hat{T}_{uv}^\dagger(1,-1)|SS\rangle \sqrt{\frac{2S-1}{2S+1}}
\end{aligned}$$

where $u \neq v$, the first term is

$$\begin{aligned}
\hat{T}_{uv}^\dagger(1,1)(\hat{S}_-)^2|SS\rangle &= -\hat{u}_\alpha^\dagger\hat{v}_\beta \sum_{xy} \hat{x}_\beta^\dagger\hat{x}_\alpha\hat{y}_\beta^\dagger\hat{y}_\alpha|\Phi_0\rangle \\
&= -\sum_{xy}(\hat{u}_\alpha^\dagger\delta_{vx}\hat{x}_\alpha\hat{y}_\beta^\dagger\hat{y}_\alpha + \hat{u}_\alpha^\dagger\hat{x}_\beta^\dagger\hat{x}_\alpha\delta_{vy}\hat{y}_\alpha)|\Phi_0\rangle \\
&= -\sum_{xy}(\delta_{vx}\delta_{ux}\hat{y}_\beta^\dagger\hat{y}_\alpha + \delta_{vx}\delta_{uy}\hat{x}_\alpha\hat{y}_\beta^\dagger - \delta_{vy}\delta_{ux}\hat{x}_\beta^\dagger\hat{y}_\alpha + \delta_{vy}\delta_{uy}\hat{x}_\beta^\dagger\hat{x}_\alpha)|\Phi_0\rangle \\
&= -(\hat{v}_\alpha\hat{u}_\beta^\dagger - \hat{u}_\beta^\dagger\hat{v}_\alpha)|\Phi_0\rangle = 2|\Phi_v^{\bar{u}}\rangle \quad (60)
\end{aligned}$$

the second term

$$\hat{T}_{uv}^\dagger(1,0)\hat{S}_-|SS\rangle = \frac{1}{\sqrt{2}}(\hat{u}_\alpha^\dagger\hat{v}_\alpha - \hat{u}_\beta^\dagger\hat{v}_\beta) \sum_x \hat{x}_\beta^\dagger\hat{x}_\alpha|\Phi_0\rangle$$

$$= \frac{1}{\sqrt{2}}(-|\Phi_v^{\bar{u}}\rangle - |\Phi_v^{\bar{u}}\rangle) = -\frac{1}{\sqrt{2}} \times 2|\Phi_v^{\bar{u}}\rangle \quad (61)$$

and the third term

$$\hat{T}_{uv}^\dagger(1, -1)|SS\rangle = \hat{u}_\beta^\dagger \hat{v}_\alpha |\Phi_0\rangle = |\Phi_v^{\bar{u}}\rangle \quad (62)$$

finally, this excited state becomes

$$\begin{aligned} [\hat{T}_{uv}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \left(\frac{1}{S\sqrt{(2S+1)(2S-1)}} + \frac{1}{S}\sqrt{\frac{2S-1}{2S+1}} + \sqrt{\frac{2S-1}{2S+1}} \right) |\Phi_v^{\bar{u}}\rangle \\ &= \sqrt{\frac{2S+1}{2S-1}} |\Phi_v^{\bar{u}}\rangle \end{aligned} \quad (63)$$

after normalized, this state becomes

$$\boxed{|\Phi_{uv}^{(S-1)(S-1)}\rangle = |\Phi_v^{\bar{u}}\rangle} \quad (64)$$

e. $\hat{T}_{O_1O_1}^\dagger(1), |\Phi_u^{(S-1)(S-1)}\rangle$ **1. The excited state formulation:**

$$\begin{aligned} [\hat{T}_{O_1O_1}^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= \hat{T}_{uu}^\dagger(1, 1)(\hat{S}_-)^2|SS\rangle \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \\ &\quad + \hat{T}_{uu}^\dagger(1, 0)\hat{S}_-|S_i S_i\rangle \left(-\frac{1}{S}\sqrt{\frac{2S-1}{2(2S+1)}} \right) \\ &\quad + \hat{T}_{uu}^\dagger(1, -1)|SS\rangle \sqrt{\frac{2S-1}{2S+1}} \end{aligned}$$

the first term is

$$\begin{aligned} \hat{T}_{uu}^\dagger(1, 1)(\hat{S}_-)^2|SS\rangle &= -\hat{u}_\alpha^\dagger \hat{u}_\beta \sum_{x \neq y} \hat{x}_\beta^\dagger \hat{x}_\alpha \hat{y}_\beta^\dagger \hat{y}_\alpha |\Phi_0\rangle \\ &= -\sum_{x \neq y} (\delta_{ux} \delta_{ux} \hat{y}_\beta^\dagger \hat{y}_\alpha + \delta_{ux} \delta_{uy} \hat{x}_\alpha \hat{y}_\beta^\dagger - \delta_{uy} \delta_{ux} \hat{x}_\beta \hat{y}_\alpha + \delta_{uy} \delta_{uy} \hat{x}_\beta^\dagger \hat{x}_\alpha) |\Phi_0\rangle \\ &= -\sum_{x \neq y} (2\delta_{uy}^2 \hat{x}_\beta^\dagger \hat{x}_\alpha - 2\delta_{ux} \delta_{uy} \hat{u}_\beta^\dagger \hat{u}_\alpha) |\Phi_0\rangle \\ &= -2 \sum_{x \neq u} \hat{x}_\beta^\dagger \hat{x}_\alpha |\Phi_0\rangle = -2 \sum_{x \neq u} |\Phi_x^{\bar{x}}\rangle \end{aligned} \quad (65)$$

the second term is

$$\begin{aligned}
\hat{T}_{uv}^\dagger(1,0)\hat{S}_-|SS\rangle &= \frac{1}{\sqrt{2}}(\hat{u}_\alpha^\dagger \hat{u}_\alpha - \hat{u}_\beta^\dagger \hat{u}_\beta) \sum_x \hat{x}_\beta^\dagger \hat{x}_\alpha |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}}(\sum_{x \neq u} \hat{x}_\beta^\dagger \hat{x}_\alpha - \sum_{x \neq u} \hat{u}_\alpha \hat{u}_\alpha^\dagger \hat{x}_\beta^\dagger \hat{x}_\alpha - \sum_x \hat{u}_\beta^\dagger \delta_{xu} \hat{x}_\alpha) |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}}(\sum_{x \neq u} \hat{x}_\beta^\dagger \hat{x}_\alpha + \sum_{x \neq u} \hat{u}_\alpha \hat{x}_\beta^\dagger \delta_{xu} - \sum_x \hat{u}_\beta^\dagger \delta_{xu} \hat{x}_\alpha) |\Phi_0\rangle \\
&= \frac{1}{\sqrt{2}}(\sum_{x \neq u} |\Phi_x^{\bar{x}}\rangle - |\Phi_u^{\bar{u}}\rangle) \tag{66}
\end{aligned}$$

the third term is

$$\hat{T}_{uu}^\dagger(1,-1)|SS\rangle = \hat{u}_\beta^\dagger \hat{u}_\alpha |\Phi_0\rangle = |\Phi_u^{\bar{u}}\rangle \tag{67}$$

finally, the excited state with $O_1 O_1$ excitation becomes

$$\begin{aligned}
[\hat{T}_u^\dagger(1) \times |S\rangle]_{S-1}^{S-1} &= (-\frac{1}{2S} \frac{2}{\sqrt{(2S+1)(2S-1)}} - \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}}) \sum_{x \neq u} |\Phi_x^{\bar{x}}\rangle \\
&\quad + (\frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} + \sqrt{\frac{2S-1}{2S+1}}) |\Phi_u^{\bar{u}}\rangle \tag{68}
\end{aligned}$$

$$= -\frac{1}{2S} \sqrt{\frac{2S+1}{2S-1}} \sum_{x \neq u} |\Phi_x^{\bar{x}}\rangle + \frac{1}{2S} \sqrt{(2S+1)(2S-1)} |\Phi_u^{\bar{u}}\rangle \tag{69}$$

$$\begin{aligned}
&= \frac{1}{2S} \sqrt{\frac{2S+1}{2S-1}} \left((2S-1) |\Phi_u^{\bar{u}}\rangle - \sum_{x \neq u} |\Phi_x^{\bar{x}}\rangle \right) \\
&= \frac{1}{2S} \sqrt{\frac{2S+1}{2S-1}} \left((2S-1) |\Phi_u^{\bar{u}}\rangle - \sum_x |\Phi_x^{\bar{x}}\rangle + |\Phi_u^{\bar{u}}\rangle \right) \\
&= \frac{1}{2S} \sqrt{\frac{2S+1}{2S-1}} \left(2S |\Phi_u^{\bar{u}}\rangle - \sum_x |\Phi_x^{\bar{x}}\rangle \right) \tag{70}
\end{aligned}$$

the norm of this excited state is

$$\langle \Phi_u^{\bar{u}} | \Phi_v^{\bar{v}} \rangle = \delta_{uv} \tag{71}$$

$$\sum_{x, y \neq u} \langle \Phi_x^{\bar{x}} | \Phi_y^{\bar{y}} \rangle = 2S - 1 \tag{72}$$

$$\sum_{x \neq u} \langle \Phi_x^{\bar{x}} | \Phi_u^{\bar{u}} \rangle = 0 \tag{73}$$

then

$$\langle \Phi_u^{(S-1)(S-1)} | \Phi_v^{(S-1)(S-1)} \rangle = \frac{(2S+1)(2S-1)}{(2S)^2} \delta_{uv} + \frac{2S-1}{(2S)^2} \frac{2S+1}{2S-1} \quad (74)$$

if $u = v$, then

$$\langle \Phi_u^{(S-1)(S-1)} | \Phi_u^{(S-1)(S-1)} \rangle = \frac{(2S+1)(2S-1)}{(2S)^2} + \frac{2S-1}{(2S)^2} \frac{2S+1}{2S-1} = \frac{2S+1}{2S} \quad (75)$$

after normalized, this state becomes

$$|\Phi_u^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}} |\Phi_u^{\bar{u}}\rangle - \frac{1}{\sqrt{2S(2S-1)}} \sum_{x \neq u} |\Phi_x^{\bar{x}}\rangle \quad (76)$$

2. Orthonormalized excited state: Consider a set of $N = 2S$ basis vector

$$\{|\Phi_{x_1}^{\bar{x}_1}\rangle, |\Phi_{x_2}^{\bar{x}_2}\rangle, \dots, |\Phi_{x_N}^{\bar{x}_N}\rangle\} \quad (77)$$

then a new set of spin-adapted state can be constructed by using the above configurations, which is $|V_l\rangle = \sum_n V_{nl} |\Phi_{x_n}^{\bar{x}_n}\rangle$ for fixed l .

$$|V_0\rangle = \frac{1}{\sqrt{N}} (|\Phi_{x_1}^{\bar{x}_1}\rangle + |\Phi_{x_2}^{\bar{x}_2}\rangle) + \dots |\Phi_{x_N}^{\bar{x}_N}\rangle \quad (78)$$

$$|V_1\rangle = \frac{N-1}{\sqrt{N(N-1)}} |\Phi_{x_1}^{\bar{x}_1}\rangle - \frac{1}{\sqrt{N(N-1)}} (|\Phi_{x_2}^{\bar{x}_2}\rangle) + \dots |\Phi_{x_N}^{\bar{x}_N}\rangle \quad (79)$$

...

$$|V_N\rangle = \frac{1}{\sqrt{2}} (|\Phi_{x_{N-1}}^{\bar{x}_{N-1}}\rangle + |\Phi_{x_N}^{\bar{x}_N}\rangle) \quad (80)$$

then the coefficients construct a matrix

$$(V_{kl}) = \begin{bmatrix} \frac{N-1}{\sqrt{N(N-1)}} & 0 & \dots & 0 \\ -\frac{1}{\sqrt{N(N-1)}} & -\frac{N-2}{\sqrt{(N-1)(N-2)}} & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ -\frac{1}{\sqrt{N(N-1)}} & -\frac{1}{\sqrt{(N-1)(N-2)}} & \dots & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{N(N-1)}} & -\frac{1}{\sqrt{(N-1)(N-2)}} & \dots & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad (81)$$

then a funny property (comes from the way constructing this matrix) will be found, which is

$$\sum_k V_{kl} \equiv 0, \quad l \geq 2 \quad (82)$$

then, construct a new excited state

$$\begin{aligned} |\Phi_v^{(S-1)(S-1)}\rangle &= \sqrt{\frac{2S-1}{2S+1}} \sum_u [\hat{T}_u^\dagger(1) \times |S\rangle]_{S-1}^{S-1} V_{uv} \\ &= \frac{1}{2S} \sum_u (2S|\Phi_u^{\bar{u}}\rangle - \sum_x |\Phi_x^{\bar{x}}\rangle) V_{uv} \end{aligned} \quad (83)$$

$$\begin{aligned} &= \sum_u V_{uv} |\Phi_u^{\bar{u}}\rangle - \sum_u |V_0\rangle V_{uv} \\ &= |V_v\rangle - |V_0\rangle \sum_u V_{uv} \equiv \boxed{|V_v\rangle} \end{aligned} \quad (84)$$

and then

$$\langle \Phi_u^{(S-1)(S-1)} | \Phi_v^{(S-1)(S-1)} \rangle \equiv \delta_{uv} \quad (85)$$

which means this kind of excited state is orthonormalized.

3. Transformation of XSF-TDA result: In the XSF-TDA calculation, we use the excited state formulated as the above:

$$|\Phi_u\rangle' = \sum_v |\Phi_v\rangle V_{vu} \quad (86)$$

$$|\Phi_v\rangle = \sqrt{\frac{2S-1}{2S+1}} [\hat{T}_v^\dagger(1) \times |S\rangle]_{S-1}^{S-1} = \boxed{|\Phi_v^{\bar{v}}\rangle - \frac{1}{2S} \sum_x |\Phi_x^{\bar{x}}\rangle} \quad (87)$$

and then, after XSF-TDA performed, we need to transfer this basis to the origin basis, now we have

$$|\Psi_I\rangle = \sum_u X_{uI} |\Phi_u\rangle' = \sum_u \sum_v X_{uI} |\Phi_v\rangle V_{vu} = \sum_k \left(\overbrace{\sum_u V_{vu} X_{uI}}^{X'_{vI}} \right) |\Phi_v\rangle \quad (88)$$

it is means that we can left multiply the matrix V_{vu} to the CI coefficient X_{uI} to get the CI coefficient corresponding to the basis of $\{|\Phi_v\rangle\}$ instead of $\{|\Phi_u\rangle'\}$.

4. About the constant term: The formulation of Eq. (87) can be further

simplified, i.e. the second term can be neglected: the matrix elements involving the state $|\Phi_u^{(S-1)(S-1)}\rangle$ can be written as

$$\langle \Psi_I^{S'S'} | \hat{h}^{\text{SO}} | \Psi_J^{(S-1)(S-1)} \rangle = \sum_{LK} X_{LI} \langle \Phi_L^{S'S'} | \hat{h}^{\text{SO}} | \Phi_K^{(S-1)(S-1)} \rangle X_{KJ} \quad (89)$$

$$= \sum_{LK} X_{LI} \langle \Phi_L^{S'S'} | \hat{h}^{\text{SO}} | \tilde{\Phi}_K^{(S-1)(S-1)} \rangle \tilde{X}_{xJ} - \sum_{Lu} \frac{1}{2S} \sum_x X_{LI} \langle \Phi_L^{S'S'} | \hat{h}^{\text{SO}} | \Phi_x^{\bar{x}} \rangle X_{uJ} \quad (90)$$

here $|\tilde{\Phi}_K^{(S-1)(S-1)}\rangle$ represent that replace $|\Phi_u^{(S-1)(S-1)}\rangle$ component by only $|\Phi_u^{\bar{u}}\rangle$. And X_{uJ} represents that the $O_1 O_1$ component of X_{KJ} . The ket of last term of Eq. (90) reads

$$\frac{1}{2S} \sum_L \sum_x X_{LI} \langle \Phi_L^{S'S'} | \hat{h}^{\text{SO}} | \Phi_x^{\bar{x}} \rangle \left(\sum_u X_{uJ} \right) \quad (91)$$

the term in bracket is equal to zero, because

$$\sum_u X_{uJ} = \sum_u \sum_v V_{uv} X_{vJ} = \sum_v X_{vJ} \left(\sum_u V_{uv} \right) = 0 \quad (92)$$

Therefore, setting

$$\boxed{|\Phi_u^{(S-1)(S-1)}\rangle = |\Phi_u^{\bar{u}}\rangle} \quad (93)$$

has no effect on the SOC matrix elements. However, in the following derivation we still employ the excited state defined in Eq. (87).

D. Summary of excited states in state interaction

Here eleven CSFs are copied from Section II C:

- $|\Phi_{ai}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S+1}} |\Phi_i^{\bar{a}}\rangle - \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} \sum_x (|\Phi_{ix}^{a\bar{x}}\rangle - |\Phi_{ix}^{\bar{a}\bar{x}}\rangle) - \frac{1}{2S\sqrt{(2S+1)(2S-1)}} \sum_{x \neq y} |\Phi_{ixy}^{a\bar{x}\bar{y}}\rangle$
- $|\Phi_{ui}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}} |\Phi_i^{\bar{u}}\rangle + \frac{1}{\sqrt{2S(2S-1)}} \sum_{x \neq u} |\Phi_{ix}^{\bar{u}\bar{x}}\rangle$
- $|\Phi_{au}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}} |\Phi_u^{\bar{a}}\rangle - \frac{1}{\sqrt{2S(2S-1)}} \sum_{x \neq u} |\Phi_{ux}^{a\bar{x}}\rangle$

- $|\Phi_{uv}^{(S-1)(S-1)}\rangle = |\Phi_v^{\bar{u}}\rangle, \quad (u \neq v)$
- $|\Phi_u^{(S-1)(S-1)}\rangle = |\Phi_u^{\bar{u}}\rangle - \frac{1}{2S} \sum_x |\Phi_x^{\bar{x}}\rangle$
- $|\Phi_0^{SS}\rangle$
- $|\Phi_{ai,1}^{SS}\rangle = \frac{1}{\sqrt{2}}(|\Phi_i^a\rangle + |\Phi_{\bar{i}}^{\bar{a}}\rangle)$
- $|\Phi_{ui}^{SS}\rangle = |\Phi_{\bar{i}}^{\bar{u}}\rangle$
- $|\Phi_{au}^{SS}\rangle = |\Phi_u^a\rangle$
- $|\Phi_{ai,2}^{SS}\rangle = \sqrt{\frac{S}{2(1+S)}}(-|\Phi_i^a\rangle + |\Phi_{\bar{i}}^{\bar{a}}\rangle - \frac{1}{S} \sum_u |\Phi_{iu}^{a\bar{u}}\rangle)$
- $|\Phi_{ai}^{(S+1)(S+1)}\rangle = -|\Phi_{\bar{i}}^a\rangle$

III. DERIVATION OF SOC MATRIX ELEMENT

A. Detailed derivation matrix elements of SOC operator

The target is to obtain this matrix element

$$\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle = \sum_{pq} h_{pq}^m \langle \Phi_K^{SS} | \hat{T}_{pq}^\dagger(1, S - S') | \Phi_L^{S'S'} \rangle. \quad (94)$$

Note that the operator $\hat{h}^m$ depends on three parameters, m , S , and S' , where m is independent of the bra and ket, while S and S' are depend on them.

B. $|\Phi^{(S-1)(S-1)}\rangle$ interact with $|\Phi^{SS}\rangle$

It is observes to see that the number of the extra matrix elements is $61 - 21 = 40 = 10 + 9 + 8 + 7 + 6$.

1. $\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj}^{(S-1)(S-1)} \rangle$ This is the most complex matrix element in those 61 elements, let us focus on the non-diagonal term first, like

$$\sum_{pq} \sum_{xy} \langle \Phi_i^{\bar{a}} | h_{pq}^m (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{jyz}^{b\bar{y}\bar{z}} \rangle \quad (95)$$

this kind of term is equal to 0 obviously, and the matrix elements below

$$\sum_{pq} \sum_x \langle \Phi_i^{\bar{a}} | h_{pq}^m (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{jx}^{b\bar{x}} \rangle \quad (96)$$

$$\sum_{pq} \sum_x \langle \Phi_i^{\bar{a}} | h_{pq}^m (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{jx}^{\bar{b}\bar{x}} \rangle \quad (97)$$

$$\sum_{pq} \sum_{x,y \neq z} \langle \Phi_{ix}^{\bar{a}\bar{x}} | h_{pq}^m (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{jyz}^{b\bar{y}\bar{z}} \rangle \quad (98)$$

$$\sum_{pq} \sum_{x,y \neq z} \langle \Phi_{ix}^{\bar{a}\bar{x}} | h_{pq}^m (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{jyz}^{b\bar{y}\bar{z}} \rangle \quad (99)$$

are also equal to zero, because the core and virtual operators can not be contracted by operator $\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta$ in the middle of matrix element. The only one that can be nonzero is

$$\begin{aligned} & \sum_{pq} \sum_{xy} \langle \Phi_{ix}^{\bar{a}\bar{x}} | h_{pq}^m (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{jy}^{b\bar{y}} \rangle \\ &= - \sum_{pq} \sum_{xy} h_{pq}^m \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{a}_\beta \} \{ \hat{p}_\beta^\dagger \hat{q}_\beta \} \{ \hat{b}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{y}_\beta \hat{j}_\alpha \} | \Phi_0 \rangle = 0 \end{aligned} \quad (100)$$

Now consider the diagonal elements

- $\langle \Phi_i^{\bar{a}} | \hat{h}^m | \Phi_j^{\bar{b}} \rangle = -\frac{1}{\sqrt{2}} (h_{ji}^m \delta_{ab} + h_{ab}^m \delta_{ij})$
- $\sum_{xy} \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{h}^m | \Phi_{jy}^{b\bar{y}} \rangle = \frac{1}{\sqrt{2}} (h_{ab}^m \delta_{xy} \delta_{ij} - h_{ji}^m \delta_{xy} \delta_{ab})$

where this term could be expanded as

$$\frac{1}{\sqrt{2}} \sum_{pq} \sum_{xy} \langle \Phi_0 | \{ \hat{i}_\alpha^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\alpha^\dagger \hat{q}_\alpha \} \{ \hat{b}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha \hat{j}_\alpha \} | \Phi_0 \rangle \quad (101)$$

notice that the $\hat{p}_\sigma^\dagger \hat{q}_\sigma$ could only be $\hat{p}_\alpha^\dagger \hat{q}_\alpha$ to make sure $\sum_{xy} \delta_{xy} h_{xy}^m$ not appear. then $\hat{p}_\alpha^\dagger \hat{q}_\alpha$ can contract with ab or ij , the latter contraction will give a minus sign.

- $\sum_{xy} \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{h}^m | \Phi_{jy}^{\bar{b}\bar{y}} \rangle = -\frac{1}{\sqrt{2}} (h_{ab}^m \delta_{xy} \delta_{ij} - h_{ji}^m \delta_{xy} \delta_{ab})$
- $\sum_{\substack{x \neq y \\ z \neq w}} \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{h}^m | \Phi_{jzw}^{b\bar{z}\bar{w}} \rangle = \frac{1}{\sqrt{2}} \sum_{xyzw} \langle \Phi_0 | \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{x}_\beta \hat{a}_\alpha (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{w}_\beta^\dagger \hat{w}_\alpha \hat{z}_\alpha \hat{j}_\beta | \Phi_0 \rangle$
the first term $\hat{p}_\alpha^\dagger \hat{q}_\alpha$ gives

$$\begin{aligned} & \sum_{xyzw} \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\alpha^\dagger \hat{q}_\alpha \} \{ \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{w}_\beta^\dagger \hat{w}_\alpha \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \\ &= h_{ab}^m \delta_{ij} \sum_{\substack{x \neq y \\ z \neq w}} (\delta_{xz} \delta_{yw} + \delta_{xw} \delta_{yz}) = 2(2S)(2S-1) h_{ab}^m \delta_{ij} \end{aligned} \quad (102)$$

and the other one will gives

$$\begin{aligned} & - \sum_{xyzw} \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\beta^\dagger \hat{q}_\beta \} \{ \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{w}_\beta^\dagger \hat{w}_\alpha \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \\ &= h_{ji}^m \delta_{ab} \sum_{\substack{x \neq y \\ z \neq w}} (\delta_{xz} \delta_{yw} + \delta_{xw} \delta_{yz}) = 2(2S)(2S-1) h_{ji}^m \delta_{ab} \end{aligned} \quad (103)$$

finally, this term give $\frac{1}{\sqrt{2}} 2(2S)(2S-1) (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab})$

So, the number of nonzero term is only two, which gives

$$\begin{aligned} & \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj}^{(S-1)(S-1)} \rangle \\ &= \frac{1}{\sqrt{2}} \left(\frac{2(2S)(2S-1)}{(2S)^2(2S+1)(2S-1)} - \frac{2S-1}{2S+1} \right) (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab}) \\ &= \frac{1}{\sqrt{2}} \left(\frac{1}{S(2S+1)} - \frac{2S-1}{2S+1} \right) (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab}) \\ &= \frac{1}{\sqrt{2}} \left(\frac{-2S^2 + S + 1}{S(2S+1)} \right) (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab}) = \frac{1-S}{S\sqrt{2}} (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab}) \end{aligned} \quad (104)$$

2. $\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{ij}^{(S-1)(S-1)} \rangle$ this matrix elements can be divided to seven terms

$$(a) \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_j^{\bar{t}} \rangle = -h_{at}^m \delta_{ij}$$

(b) $\sum_{pqx} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_j^{\bar{t}} \rangle = \sum_{pqx} h_{pq}^m \langle \Phi_0 | \{ \hat{i}_\alpha^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\alpha^\dagger \hat{q}_\alpha \} \{ \hat{t}_\beta^\dagger \hat{j}_\alpha \} | \Phi_0 \rangle = -h_{at}^m \delta_{ij}$ where notice that $\hat{a}_\alpha$ must contract $\hat{p}_\alpha^\dagger$, and ij contracted, two β contracted.

(c) $\sum_{pqx} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_j^{\bar{t}} \rangle = 0$

(d) $\sum_{pq} \sum_{x \neq t} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jx}^{\bar{t}} \rangle = 0$

(e) $\sum_{pqx} \sum_{y \neq t} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jy}^{\bar{t}\bar{y}} \rangle = 0$

(f) $\sum_{pqx} \sum_{y \neq t} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jy}^{\bar{t}\bar{y}} \rangle$

$$\begin{aligned}
&= - \sum_{pqx} \sum_{y \neq t} h_{pq}^m (\langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{a}_\beta \} \{ \hat{p}_\beta^\dagger \hat{q}_\beta \} \{ \hat{t}_\beta^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \\
&\quad + \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{a}_\beta \} \{ \hat{p}_\beta^\dagger \hat{q}_\beta \} \{ \hat{t}_\beta^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle) \\
&= \sum_x \sum_{y \neq t} h_{xy}^m \delta_{xt} \delta_{ij} \delta_{xy} - \sum_x \sum_{y \neq t} h_{at}^m \delta_{ij} \delta_{xy} = (1 - 2S) h_{at}^m \delta_{ij} \quad (105)
\end{aligned}$$

(g) $\sum_{pq} \sum_{\substack{z \neq t \\ x \neq y}} h_{pq}^m \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jz}^{\bar{t}\bar{z}} \rangle$ this matrix element can be considered as

i. $\hat{a}_\alpha$ must be contracted, then only $\hat{p}_\alpha^\dagger \hat{q}_\alpha$ term is nonvanishing, and $\hat{p}_\alpha^\dagger$ contract $\hat{a}_\alpha$.

ii. Contract ij .

then this term becomes

$$\begin{aligned}
&\sum_{pq} \sum_{\substack{z \neq t \\ x \neq y}} h_{pq}^m \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jz}^{\bar{t}\bar{z}} \rangle \\
&= \sum_{pq} \sum_{\substack{z \neq t \\ x \neq y}} h_{pq}^m \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{y}_\beta \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\alpha^\dagger \hat{q}_\alpha \} \{ \hat{t}_\beta^\dagger \hat{z}_\beta^\dagger \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle
\end{aligned}$$

$$\begin{aligned}
& + \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\beta^\dagger \hat{y}_\beta \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\alpha^\dagger \hat{q}_\alpha \} \{ \hat{t}_\beta^\dagger \hat{z}_\beta^\dagger \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \\
& = \sum_{x \neq y} \sum_{z \neq t} h_{ax}^m \delta_{yz} \delta_{ij} (-\delta_{xt} \delta_{yz} + \delta_{xz} \delta_{yt}) + \sum_{x \neq y} \sum_{z \neq t} h_{ay}^m \delta_{zx} \delta_{ij} (\delta_{xt} \delta_{yz} - \delta_{xz} \delta_{yt}) \\
& = -2(2S-1) \delta_{ij} h_{at}^m
\end{aligned} \tag{106}$$

Here the summation need to be treated carefully:

$$\begin{aligned}
- \sum_{x \neq y} \sum_{z \neq t} h_{ax}^m \delta_{yz} \delta_{ij} \delta_{xt} &= - \sum_{x \neq y} \left[h_{ax}^m \delta_{yz} \delta_{ij} \delta_{xt} \right] \\
&= - \sum_{x \neq y} h_{ax}^m \delta_{ij} \delta_{xt} = -(2S-1) h_{ax}^m \delta_{ij} \delta_{xt}
\end{aligned} \tag{107}$$

Finally, this matrix element becomes

$$\begin{aligned}
& \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{(S-1)(S-1)} \rangle \\
&= \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2S(2S+1)}} \left(1 - 2S + \frac{2S-1}{2S} + \frac{1}{S} + \frac{1}{2S} - 1 \right) \delta_{ij} h_{at}^m \\
&= \frac{1-S}{2S} \sqrt{\frac{2S+1}{S}} \delta_{ij} h_{at}^m
\end{aligned} \tag{108}$$

$$\begin{aligned}
3. \quad & \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{(S-1)(S-1)} \rangle \\
&= \frac{1-S}{2S} \sqrt{\frac{2S+1}{S}} \delta_{ab} h_{ti}^m
\end{aligned} \tag{109}$$

$$4. \quad \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tw}^{(S-1)(S-1)} \rangle = 0, \text{ where}$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{w}} \rangle = 0 \tag{110}$$

$$\sum_{pqx} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{w}} \rangle = \sum_x h_{ai}^m \delta_{xw} \delta_{xt} = 0 \tag{111}$$

$$\sum_{pqx} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{w}} \rangle = - \sum_x h_{ai}^m \delta_{xw} \delta_{xt} = 0 \tag{112}$$

are equal to zero because of $w \neq t$.

$$5. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = 0 \quad (113)$$

$$\sum_{pqx} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_x^{\bar{x}} \rangle = 0 \quad (114)$$

$$\sum_{pqx} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = \sum_x h_{ai}^m \delta_{xt} = h_{ai}^m \quad (115)$$

$$\sum_{pqxy} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_y^{\bar{y}} \rangle = \sum_{xy} h_{ai}^m \delta_{xy} = (2S) h_{ai}^m \quad (116)$$

$$\sum_{pqx} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = - \sum_x h_{ai}^m \delta_{xt} = -h_{ai}^m \quad (117)$$

$$\sum_{pqxy} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_y^{\bar{y}} \rangle = - \sum_x h_{ai}^m \delta_{xy} = -(2S) h_{ai}^m \quad (118)$$

then

$$\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle = 0 \quad (119)$$

$$6. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_0^{SS} \rangle$$

$$\begin{aligned} &= \sqrt{\frac{2S-1}{2S+1}} \langle \Phi_i^{\bar{a}} | \hat{h}^m | \Phi_0 \rangle \\ &= \sqrt{\frac{2S-1}{2S+1}} \sum_{pq} h_{pq}^m \langle \Phi_0 | \{ \hat{i}_\alpha^\dagger \hat{a}_\beta \} \{ \hat{p}_\beta^\dagger \hat{q}_\alpha \} | \Phi_0 \rangle = \sqrt{\frac{2S-1}{2S+1}} h_{ai}^m \end{aligned} \quad (120)$$

$$7. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = h_{ab}^m \delta_{ij} \quad (121)$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = -h_{ji}^m \delta_{ab} \quad (122)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (123)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = 0 \quad (124)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (125)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{b}} \rangle = 0 \quad (126)$$

then

$$\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle = \frac{1}{\sqrt{2}} \sqrt{\frac{2S-1}{2S+1}} (h_{ab}^m \delta_{ij} - h_{ji}^m \delta_{ab}) \quad (127)$$

$$8. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{t}} \rangle = 0 \quad (128)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{t}} \rangle = 0 \quad (129)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{t}} \rangle = -h_{at}^m \delta_{ij} \quad (130)$$

then

$$\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle = -\frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} h_{at}^m \delta_{ij} \quad (131)$$

$$9. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = 0 \quad (132)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = -h_{ti}^m \delta_{ab} \quad (133)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = 0 \quad (134)$$

then

$$\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle = \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} h_{ti}^m \delta_{ab} \quad (135)$$

$$10. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = h_{ab}^m \delta_{ij}$$

$$\begin{aligned}
\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle &= -h_{ji}^m \delta_{ab} \\
\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle &= 0 \\
\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle &= 0 \\
\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{b}} \rangle &= 0 \\
\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{b}} \rangle &= 0 \\
\sum_y \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jy}^{b\bar{y}} \rangle &= 0
\end{aligned} \tag{136}$$

$$\sum_{xy} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jy}^{b\bar{y}} \rangle = -(2S) h_{ji}^m \delta_{ab} \tag{137}$$

$$\sum_{xy} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jy}^{b\bar{y}} \rangle = (2S) h_{ab}^m \delta_{ij} \tag{138}$$

$$\sum_{x \neq y} \sum_z \sum_{pq} h_{pq}^m \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jz}^{b\bar{z}} \rangle = 0 \tag{139}$$

the last term is more difficult to deriving, which the contraction can be viewed as

$$\langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\beta^\dagger \hat{q}_\alpha \} \{ \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \tag{140}$$

notice that (a). ij, ab contracted. (b). pq, zz not contracted. (c). pq contracted with xy, yx . Then

$$\begin{aligned}
& \sum_{x \neq y} \sum_z \sum_{pq} h_{pq}^m \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jz}^{b\bar{z}} \rangle \\
&= \sum_{x \neq y} \sum_z \sum_{pq} h_{pq}^m \left(\langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\beta^\dagger \hat{q}_\alpha \} \{ \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \right. \\
&\quad \left. + \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{y}_\alpha^\dagger \hat{y}_\beta \hat{x}_\beta \hat{a}_\alpha \} \{ \hat{p}_\beta^\dagger \hat{q}_\alpha \} \{ \hat{b}_\alpha^\dagger \hat{z}_\beta^\dagger \hat{z}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \right)
\end{aligned}$$

$$= - \sum_{x \neq y} \sum_z (h_{xy}^m \delta_{ab} \delta_{ij} \delta_{yz} \delta_{xz} + h_{yx}^m \delta_{ab} \delta_{ij} \delta_{xz} \delta_{yz}) = 0$$

then this term becomes

$$\begin{aligned} & \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle \\ &= - \sqrt{\frac{S}{2(1+S)}} \left(\sqrt{\frac{2S-1}{2S+1}} + \frac{1}{S} \sqrt{\frac{2S-1}{2S+1}} \right) (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab}) \\ &= - \sqrt{\frac{(1+S)(2S-1)}{2S(2S+1)}} (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab}) \end{aligned} \quad (141)$$

$$11. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_j^{\bar{t}} \rangle = -(h_{ut}^m \delta_{ij} + h_{ji}^m \delta_{ut}) \quad (142)$$

$$\sum_{x \neq t} \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jx}^{\bar{t}\bar{x}} \rangle = 0 \quad (143)$$

$$\begin{aligned} & \sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jy}^{\bar{t}\bar{y}} \rangle = (2S-1) h_{ji}^m \delta_{ut} + (3-2S) h_{ut}^m \delta_{ij} \\ & \quad (144) \end{aligned}$$

here the last term is complex, which can expanded as

$$\begin{aligned} & \sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jy}^{\bar{t}\bar{y}} \rangle \\ &= \sum_{x \neq u} \sum_{pq} h_{pq}^m \left(\langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{u}_\beta \} \{ \hat{p}_\alpha^\dagger \hat{q}_\alpha \} \{ \hat{t}_\beta^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \right. \\ & \quad - \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{u}_\beta \} \{ \hat{p}_\beta^\dagger \hat{q}_\beta \} \{ \hat{t}_\beta^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \\ & \quad \left. - \langle \Phi_0 | \{ \hat{i}_\beta^\dagger \hat{x}_\alpha^\dagger \hat{x}_\beta \hat{u}_\beta \} \{ \hat{p}_\beta^\dagger \hat{q}_\beta \} \{ \hat{t}_\beta^\dagger \hat{y}_\beta^\dagger \hat{y}_\alpha \hat{j}_\beta \} | \Phi_0 \rangle \right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{\substack{x \neq u \\ y \neq t}} h_{yx}^m \delta_{xt} \delta_{uy} \delta_{ij} - \sum_{\substack{x \neq u \\ y \neq t}} h_{ji}^m \delta_{xy} (-\delta_{xy} \delta_{ut} + \cancel{\delta_{xy} \delta_{xt}}) \\
&\quad - \sum_{\substack{x \neq u \\ y \neq t}} (h_{ut}^m \delta_{xy} - \cancel{h_{xy}^m \delta_{xt}} - \cancel{h_{xt}^m \delta_{uy}} + \cancel{h_{xy}^m \delta_{ut}}) \delta_{xy} \delta_{ij} \\
&= h_{ut}^m \delta_{ij} + (2S-1) h_{ji}^m \delta_{ut} - (2S-2) h_{ut}^m \delta_{ij} \\
&= (2S-1) h_{ji}^m \delta_{ut} + (3-2S) h_{ut}^m \delta_{ij}
\end{aligned}$$

a very important thing to noticed here is $\sum_{\substack{x \neq u \\ y \neq t}} h_{ut}^m \delta_{xy} = (2S-2)$, because $u \neq x = y \neq t$, x and y can takes $2S-2$ values. Then the finally result becomes

$$\begin{aligned}
&\langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{(S-1)(S-1)} \rangle \\
&= \frac{1}{\sqrt{2}} \left[-\frac{2S-1}{2S} (h_{ut}^m \delta_{ij} + h_{ji}^m \delta_{ut}) + \frac{1}{2S} h_{ji}^m \delta_{ut} + \frac{3-2S}{2S(2S-1)} h_{ut}^m \delta_{ij} \right] \quad (145)
\end{aligned}$$

$$\begin{aligned}
&= -\frac{S-1}{S\sqrt{2}} h_{ji}^m \delta_{ut} - \frac{(S-1)(2S+1)}{\sqrt{2}S(2S-1)} h_{ut}^m \delta_{ij} \\
&= -\frac{S-1}{S\sqrt{2}} (h_{ji}^m \delta_{ut} + \frac{2S+1}{2S-1} h_{ut}^m \delta_{ij}) \quad (146)
\end{aligned}$$

$$12. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{(S-1)(S-1)} \rangle = 0$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_v^{\bar{b}} \rangle = 0 \quad (147)$$

$$\sum_{x \neq v} \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{vx}^{b\bar{x}} \rangle = 0 \quad (148)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{u\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_v^{\bar{b}} \rangle = 0 \quad (149)$$

$$\sum_{\substack{x \neq u \\ y \neq v}} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{u\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{vy}^{b\bar{y}} \rangle = 0 \quad (150)$$

$$13. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tw}^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_w^{\bar{t}} \rangle = -h_{wi}^m \delta_{ut} \quad (151)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_w^{\bar{t}} \rangle = h_{wi}^m \delta_{ut} \quad (152)$$

then

$$\begin{aligned} & \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tw}^{(S-1)(S-1)} \rangle \\ &= -\frac{1}{\sqrt{2}} \left(\sqrt{\frac{2S-1}{2S}} - \frac{1}{\sqrt{2S(2S-1)}} \right) h_{wi}^m \delta_{ut} = -\frac{S-1}{\sqrt{S(2S-1)}} h_{wi}^m \delta_{ut} \end{aligned} \quad (153)$$

$$14. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = -h_{ui}^m \delta_{ut} \quad (154)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_x^{\bar{x}} \rangle = -h_{ui}^m \quad (155)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = -(1 - \delta_{ut}) h_{ui}^m \quad (156)$$

$$\sum_{x \neq u} \sum_y \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_y^{\bar{y}} \rangle = -(2S-1) h_{ui}^m \quad (157)$$

then the final result becomes

$$\begin{aligned} & \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle \\ &= \frac{1}{\sqrt{2}} \sqrt{\frac{2S-1}{2S}} \left(-\delta_{ut} + \frac{1}{2S} \right) h_{ui}^m + \frac{1}{2\sqrt{S(2S-1)}} \left(-(1 - \delta_{ut}) + \frac{2S-1}{2S} \right) h_{ui}^m \\ &= -\frac{1}{2\sqrt{S(2S-1)}} \left(\frac{1-S}{S} + 2(S-1)\delta_{ut} \right) h_{ui}^m \end{aligned} \quad (158)$$

here one must to be carefully to make sure the result is right:

$$\begin{aligned} & \sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle \\ &= -\sum_{x \neq u} h_{ui}^m \delta_{xt} = -\left(\sum_x h_{ui}^m \delta_{xt} - h_{ui}^m \delta_{ut} \right) = -(1 - \delta_{ut}) h_{ui}^m \end{aligned} \quad (159)$$

$$15. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_0^{SS} \rangle$$

$$= \sqrt{\frac{2S-1}{2S}} h_{ui}^m \quad (160)$$

$$16. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = h_{ub}^m \delta_{ij} \quad (161)$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = 0 \quad (162)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (163)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = 0 \quad (164)$$

then

$$\langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle = \frac{1}{2} \sqrt{\frac{2S-1}{S}} h_{ub}^m \delta_{ij} \quad (165)$$

$$17. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = -h_{ji}^m \delta_{ut} \quad (166)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = -h_{ut}^m \delta_{ij} \quad (167)$$

then

$$\begin{aligned} & \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle \\ &= -\sqrt{\frac{2S-1}{2S}} h_{ji}^m \delta_{ut} - \frac{1}{\sqrt{2S(2S-1)}} h_{ut}^m \delta_{ij} \\ &= -\sqrt{\frac{2S-1}{2S}} \left(h_{ji}^m \delta_{ut} + \frac{1}{2S-1} h_{ut}^m \delta_{ij} \right) \end{aligned} \quad (168)$$

$$18. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle = 0$$

$$\langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = 0 \quad (169)$$

$$\sum_{x \neq u} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = 0 \quad (170)$$

$$19. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = h_{ub}^m \delta_{ij} \quad (171)$$

$$\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = 0 \quad (172)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jx}^{b\bar{x}} \rangle = 0 \quad (173)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (174)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{b}} \rangle = 0 \quad (175)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jx}^{b\bar{x}} \rangle = \sum_{xy} h_{ub}^m \delta_{xy} \delta_{ij} = (2S-1) h_{ub}^m \delta_{ij} \quad (176)$$

then the final result becomes

$$\begin{aligned} & \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle \\ &= \sqrt{\frac{S}{2(1+S)}} \left(-\sqrt{\frac{2S-1}{2S}} - \frac{1}{\sqrt{2S(2S-1)}} \frac{2S-1}{S} \right) h_{ub}^m \delta_{ij} \\ &= -\frac{\sqrt{(S+1)(2S-1)}}{2S} h_{ub}^m \delta_{ij} \end{aligned} \quad (177)$$

$$20. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{b}} \rangle = -(h_{tu}^m \delta_{ab} + h_{ab}^m \delta_{ut}) \quad (178)$$

$$\sum_{x \neq t} \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{ty}^{b\bar{y}} \rangle = 0 \quad (179)$$

$$\sum_{\substack{x \neq u \\ y \neq t}} \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{ty}^{b\bar{y}} \rangle = (2S-1) h_{ab}^m \delta_{ut} + (3-2S) h_{tu}^m \delta_{ab} \quad (180)$$

then

$$\langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{(S-1)(S-1)} \rangle = -\frac{S-1}{S\sqrt{2}} (h_{ab}^m \delta_{ut} + \frac{2S+1}{2S-1} h_{tu}^m \delta_{ab}) \quad (181)$$

$$21. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tw}^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_w^{\bar{t}} \rangle = -h_{at}^m \delta_{uw} \quad (182)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_w^{\bar{t}} \rangle = -h_{at}^m \delta_{uw} \quad (183)$$

then

$$\begin{aligned} & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tw}^{(S-1)(S-1)} \rangle \\ &= \frac{1}{\sqrt{2}} \left(-\sqrt{\frac{2S-1}{2S}} + \frac{1}{\sqrt{2S(2S-1)}} \right) h_{at}^m \delta_{uw} = \frac{1-S}{\sqrt{S(2S-1)}} h_{at}^m \delta_{uw} \quad (184) \end{aligned}$$

$$22. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = -h_{au}^m \delta_{ut} \quad (185)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_x^{\bar{x}} \rangle = -h_{au}^m \quad (186)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = (1 - \delta_{ut}) h_{au}^m \quad (187)$$

$$\sum_{x \neq u} \sum_y \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_y^{\bar{y}} \rangle = (2S-1) h_{au}^m \quad (188)$$

then

$$\begin{aligned} & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle \\ &= \frac{1}{\sqrt{2}} \sqrt{\frac{2S-1}{2S}} \left(-\delta_{ut} + \frac{1}{2S} \right) h_{au}^m - \frac{1}{2\sqrt{S(2S-1)}} \left((1 - \delta_{ut}) - \frac{2S-1}{2S} \right) h_{au}^m \\ &= -\frac{1}{2\sqrt{S(2S-1)}} \left(\frac{1-S}{S} + 2(S-1)\delta_{ut} \right) h_{au}^m \quad (189) \end{aligned}$$

$$23. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_0^{SS} \rangle$$

$$= \sqrt{\frac{2S-1}{2S}} \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_0 \rangle = \sqrt{\frac{2S-1}{2S}} h_{au}^m \quad (190)$$

$$24. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle = -\frac{1}{2} \sqrt{\frac{2S-1}{S}} h_{ju}^m \delta_{ab}$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (191)$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = -h_{ju}^m \delta_{ab} \quad (192)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (193)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = 0 \quad (194)$$

$$25. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle = 0$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = 0 \quad (195)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = 0 \quad (196)$$

$$26. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = h_{ab}^m \delta_{ut} \quad (197)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = -h_{tu}^m \delta_{ab} \quad (198)$$

then

$$\begin{aligned} & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle \\ &= \sqrt{\frac{2S-1}{2S}} h_{ab}^m \delta_{ut} + \frac{1}{\sqrt{2S(2S-1)}} h_{tu}^m \delta_{ab} \\ &= \sqrt{\frac{2S-1}{2S}} \left(h_{ab}^m \delta_{ut} + \frac{1}{2S-1} h_{tu}^m \delta_{ab} \right) \end{aligned} \quad (199)$$

$$27. \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (200)$$

$$\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{b}} \rangle = -h_{ju}^m \delta_{ab} \quad (201)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{a}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jx}^{b\bar{x}} \rangle = 0 \quad (202)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0 \quad (203)$$

$$\sum_{x \neq u} \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{b}} \rangle = 0 \quad (204)$$

$$\sum_{x \neq u} \sum_y \sum_{pq} h_{pq}^m \langle \Phi_{ux}^{a\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{j\bar{y}}^{b\bar{y}} \rangle = - \sum_{x \neq u, y} h_{ju}^m \delta_{xy} \delta_{ab} = -(2S-1) h_{ju}^m \delta_{ab} \quad (205)$$

then

$$\begin{aligned} & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle \\ &= \sqrt{\frac{S}{2(1+S)}} \left(-\sqrt{\frac{2S-1}{2S}} - \frac{1}{\sqrt{2S(2S-1)}} \frac{2S-1}{S} \right) h_{ju}^m \delta_{ab} \\ &= -\frac{\sqrt{(S+1)(2S-1)}}{2S} h_{ju}^m \delta_{ab} \end{aligned} \quad (206)$$

$$\begin{aligned} 28. \quad & \langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tw}^{(S-1)(S-1)} \rangle \\ &= \frac{1}{\sqrt{2}} \langle \Phi_v^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_w^{\bar{t}} \rangle = -\frac{1}{\sqrt{2}} (h_{wv}^m \delta_{ut} + h_{ut}^m \delta_{vw}) \end{aligned} \quad (207)$$

$$\begin{aligned} 29. \quad & \langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle \\ & \sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = -(h_{tv}^m \delta_{ut} + h_{ut}^m \delta_{vt}) \end{aligned} \quad (208)$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_x^{\bar{x}} \rangle = -2h_{uv}^m \quad (209)$$

then this term becomes

$$\begin{aligned} & \langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle \\ &= -\frac{1}{\sqrt{2}} (h_{tv}^m \delta_{ut} + h_{ut}^m \delta_{vt}) + \frac{2}{2S\sqrt{2}} h_{uv}^m \\ &= -\frac{1}{\sqrt{2}} (h_{tv}^m \delta_{ut} + h_{ut}^m \delta_{vt} - \frac{1}{S} h_{uv}^m) \end{aligned} \quad (210)$$

$$\begin{aligned}
&= -\frac{1}{\sqrt{2}}(h_{uv}^m \delta_{ut} + h_{uv}^m \delta_{vt} - \frac{1}{S} h_{uv}^m) \\
&= -\frac{1}{\sqrt{2}}(h_{uv}^m \delta_{ut} + h_{uv}^m (\delta_{vt} - \frac{1}{S}))
\end{aligned} \tag{211}$$

$$\begin{aligned}
30. \quad &\langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_0^{SS} \rangle \\
&= \sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_0 \rangle = h_{uv}^m
\end{aligned} \tag{212}$$

$$\begin{aligned}
31. \quad &\langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle = 0 \\
&\sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0
\end{aligned} \tag{213}$$

$$\sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{\bar{j}}^{\bar{b}} \rangle = 0 \tag{214}$$

$$\begin{aligned}
32. \quad &\langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle \\
&= \sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = -h_{jv}^m \delta_{tu}
\end{aligned} \tag{215}$$

$$\begin{aligned}
33. \quad &\langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle \\
&= \sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = h_{ub}^m \delta_{vt}
\end{aligned} \tag{216}$$

$$\begin{aligned}
34. \quad &\langle \Phi_{uv}^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle \\
&= \sum_x \sum_{pq} h_{pq}^m \langle \Phi_v^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jx}^{b\bar{x}} \rangle = 0
\end{aligned} \tag{217}$$

$$\begin{aligned}
35. \quad &\langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_t^{(S-1)(S-1)} \rangle = 0 \\
&\sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_t^{\bar{t}} \rangle = 0
\end{aligned} \tag{218}$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_x^{\bar{x}} \rangle = 0 \tag{219}$$

$$\sum_{xy} \sum_{pq} h_{pq}^m \langle \Phi_x^{\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_y^{\bar{y}} \rangle = 0 \tag{220}$$

$$\begin{aligned}
36. \quad & \langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_0^{SS} \rangle \\
& = \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_0 \rangle = 0
\end{aligned} \tag{221}$$

$$\begin{aligned}
37. \quad & \langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,1}^{SS} \rangle \\
& = \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^b \rangle = 0
\end{aligned} \tag{222}$$

$$\begin{aligned}
38. \quad & \langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle \\
& \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = -h_{ju}^m \delta_{ut}
\end{aligned} \tag{223}$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_x^{\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_j^{\bar{t}} \rangle = -h_{jt}^m \tag{224}$$

then this term becomes

$$\langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{tj}^{SS} \rangle = -(\delta_{ut} - \frac{1}{2S}) h_{jt}^m \tag{225}$$

$$\begin{aligned}
39. \quad & \langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle \\
& \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = h_{ub}^m \delta_{ut}
\end{aligned} \tag{226}$$

$$\sum_x \sum_{pq} h_{pq}^m \langle \Phi_x^{\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_t^b \rangle = h_{tb}^m \tag{227}$$

then this term becomes

$$\langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bt}^{SS} \rangle = (\delta_{ut} - \frac{1}{2S}) h_{tb}^m \tag{228}$$

$$\begin{aligned}
40. \quad & \langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle \\
& \sum_y \sum_{pq} h_{pq}^m \langle \Phi_u^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jy}^{b\bar{y}} \rangle = h_{jb}^m
\end{aligned} \tag{229}$$

$$\sum_{xy} \sum_{pq} h_{pq}^m \langle \Phi_x^{\bar{x}} | \hat{p}_\beta^\dagger \hat{q}_\alpha | \Phi_{jy}^{b\bar{y}} \rangle = (2S) h_{jb}^m \tag{230}$$

then

$$\begin{aligned} & \langle \Phi_u^{(S-1)(S-1)} | \hat{h}^m | \Phi_{bj,2}^{SS} \rangle \\ &= -\frac{1}{S} \sqrt{\frac{S}{2(1+S)}} \left(1 - \frac{2S}{2S}\right) h_{jb}^m = -\sqrt{\frac{1}{2S(1+S)}} \left(1 - \frac{2S}{2S}\right) h_{jb}^m = 0 \end{aligned} \quad (231)$$

C. $|\Phi^{SS}\rangle$ interact with $|\Phi^{(S+1)(S+1)}\rangle$

41. $\langle \Phi_0^{SS} | h^m | \Phi_0^{SS} \rangle$

$$\begin{aligned} &= \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | \hat{T}_{pq}^\dagger(1,0) | \Phi_0^{SS} \rangle \\ &= \frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_0^{SS} \rangle \equiv 0 \end{aligned} \quad (232)$$

42. $\langle \Phi_0^{SS} | h^m | \Phi_{bj,1}^{SS} \rangle$

$$\begin{aligned} &= \frac{1}{2} \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{b}_\alpha^\dagger \hat{j}_\alpha + \hat{b}_\beta^\dagger \hat{j}_\beta) | \Phi_0^{SS} \rangle \\ &= \frac{1}{2} (h_{pq}^m - h_{pq}^m) = 0 \end{aligned} \quad (233)$$

43. $\langle \Phi_0^{SS} | h^m | \Phi_{vj,1}^{SS} \rangle$

$$\begin{aligned} &= \langle \Phi_0^{SS} | h^m | \Phi_j^{\bar{v}} \rangle \\ &= \frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{v}_\beta^\dagger \hat{j}_\beta) | \Phi_0^{SS} \rangle = -\frac{1}{\sqrt{2}} h_{jv}^m \end{aligned} \quad (234)$$

44. $\langle \Phi_0^{SS} | h^m | \Phi_{bv,1}^{SS} \rangle$

$$\begin{aligned} &= \langle \Phi_0^{SS} | h^m | \Phi_a^v \rangle \\ &= \frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{b}_\alpha^\dagger \hat{v}_\alpha) | \Phi_0^{SS} \rangle = \frac{1}{\sqrt{2}} h_{vb}^m \end{aligned} \quad (235)$$

$$\begin{aligned}
45. \quad & \langle \Phi_0^{SS} | h^m | \Phi_{bj,2}^{SS} \rangle \\
&= \sqrt{\frac{S}{2(1+S)}} (-\langle \Phi_0^{SS} | h^m | \Phi_j^b \rangle + \langle \Phi_0^{SS} | h^m | \Phi_j^{\bar{b}} \rangle - \frac{1}{S} \sum_u \langle \Phi_0^{SS} | h^m | \Phi_{jv}^{b\bar{v}} \rangle) \\
&= \sqrt{\frac{S}{2(1+S)}} \left(-\frac{2}{\sqrt{2}} h_{jb}^m - \underbrace{\frac{1}{S\sqrt{2}} \sum_{pq,u} h_{pq}^m \langle \Phi_0^{SS} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{j}_\alpha^\dagger \hat{v}_\beta^\dagger \hat{v}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle}_{=0 \text{ (rank of excitations)}} \right) \\
&= -\sqrt{\frac{S}{(1+S)}} h_{jb}^m \tag{236}
\end{aligned}$$

$$\begin{aligned}
46. \quad & \langle \Phi_0^{SS} | h^m | \Phi_{bj}^{(S+1)(S+1)} \rangle \\
&= \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | \hat{T}_{pq}^\dagger(1, -1) | \Phi_j^b \rangle \\
&= -\sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{p}_\beta^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{j}_\beta) | \Phi_0^{SS} \rangle = -h_{jb}^m \tag{237}
\end{aligned}$$

$$\begin{aligned}
47. \quad & \langle \Phi_{ai,1}^{SS} | h^m | \Phi_{bj,1}^{SS} \rangle \\
&= \frac{1}{2} \left[\begin{aligned} & \sum_{pq} h_{pq}^m \langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^b \rangle \quad \rightarrow \quad \delta_{ij} h_{ab}^m - \delta_{ab} h_{ji}^m \\ & + \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{b}} \rangle \quad \rightarrow \quad -(\delta_{ij} h_{ab}^m - \delta_{ab} h_{ji}^m) \\ & + \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^b \rangle \quad \rightarrow \quad 0 \\ & + \sum_{pq} h_{pq}^m \langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{b}} \rangle \quad \rightarrow \quad 0 \end{aligned} \right] = 0 \tag{238}
\end{aligned}$$

Here

$$\begin{aligned}
& \sum_{pq} h_{pq}^m \langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha) | \Phi_j^b \rangle \\
&= \langle \Phi_0^{SS} | \overbrace{(\hat{i}_\alpha^\dagger \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha)}^{\text{}} \overbrace{(\hat{b}_\alpha^\dagger \hat{j}_\alpha)}^{\text{}} | \Phi_0^{SS} \rangle + \langle \Phi_0^{SS} | \overbrace{(\hat{i}_\alpha^\dagger \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha)}^{\text{}} \overbrace{(\hat{b}_\alpha^\dagger \hat{j}_\alpha)}^{\text{}} | \Phi_0^{SS} \rangle \\
&= \delta_{ij} h_{ab}^m - \delta_{ab} h_{ji}^m
\end{aligned}$$

$$\begin{aligned}
48. \quad & \langle \Phi_{ai,1}^{SS} | h^m | \Phi_{vj,1}^{SS} \rangle \\
&= \frac{1}{2} \sum_{pq} h_{pq}^m \left[\langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{v}} \rangle + \langle \Phi_i^{\bar{a}} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{v}} \rangle \right] \\
&= -\frac{1}{2} \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{a}} | (\hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{v}} \rangle = -\frac{1}{2} \delta_{ij} h_{av}^m \quad (239)
\end{aligned}$$

$$\begin{aligned}
49. \quad & \langle \Phi_{ai,1}^{SS} | h^m | \Phi_{bv,1}^{SS} \rangle \\
&= \frac{1}{2} \sum_{pq} h_{pq}^m \left[\langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_v^b \rangle + \langle \Phi_i^{\bar{a}} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_v^b \rangle \right] \\
&= \frac{1}{2} \sum_{pq} h_{pq}^m \langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha) | \Phi_v^b \rangle \\
&= \frac{1}{2} \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | \overbrace{(\hat{i}_\alpha^\dagger \hat{a}_\alpha)(\hat{p}_\alpha^\dagger \hat{q}_\alpha)(\hat{b}_\alpha^\dagger \hat{v}_\alpha)} | \Phi_0^{SS} \rangle = -\frac{1}{2} \delta_{ab} h_{vi}^m \quad (240)
\end{aligned}$$

$$\begin{aligned}
50. \quad & \langle \Phi_{ai,1}^{SS} | h^m | \Phi_{bj,2}^{SS} \rangle \\
&= \frac{1}{2} \sqrt{\frac{S}{2(1+S)}} \sum_{pq} h_{pq}^m \left[-\langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha) | \Phi_j^b \rangle + \langle \Phi_i^{\bar{a}} | (-\hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{b}} \rangle + R \right] \\
&= -\sqrt{\frac{S}{2(1+S)}} (\delta_{ij} h_{ab}^m - \delta_{ab} h_{ji}^m) \quad (241)
\end{aligned}$$

Here

$$R = -\frac{1}{S\sqrt{2}} \sum_u \left[\langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{ju}^{b\bar{u}} \rangle - \langle \Phi_i^{\bar{a}} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{ju}^{b\bar{u}} \rangle \right] \quad (242)$$

This term is always equal to zero, for example, consider the first term in above equation:

$$\langle \Phi_i^a | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_{ju}^{b\bar{u}} \rangle = \langle \Phi_0^{SS} | (\hat{i}_\alpha^\dagger \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{b}_\alpha^\dagger \hat{u}_\beta^\dagger \hat{u}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle \quad (243)$$

to make sure that the red term can be contracted, the term corresponding to use first operator in second bracket vanishes, and then the $\hat{i}_\alpha^\dagger \hat{u}_\alpha$ must be chosen, and $\delta_{iu} \equiv 0$.

$$\begin{aligned}
51. \quad & \langle \Phi_{ai,1}^{SS} | h^m | \Phi_{bj}^{(S+1)(S+1)} \rangle \\
&= -\frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \left[\langle \Phi_i^a | (\hat{p}_\beta^\dagger \hat{q}_\alpha) | \Phi_j^b \rangle + \langle \Phi_i^{\bar{a}} | (\hat{p}_\beta^\dagger \hat{q}_\alpha) | \Phi_j^{\bar{b}} \rangle \right] \\
&= -\frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \left[\langle \Phi_0^{SS} | (\hat{i}_\alpha^\dagger \hat{a}_\alpha) (\hat{p}_\beta^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{j}_\beta) | \Phi_0^{SS} \rangle + \langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{a}_\beta) (\hat{p}_\beta^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{j}_\beta) | \Phi_0^{SS} \rangle \right] \\
&= \frac{1}{\sqrt{2}} (\delta_{ab} h_{ji}^m - \delta_{ij} h_{ab}^m) \tag{244}
\end{aligned}$$

$$\begin{aligned}
52. \quad & \langle \Phi_{ui,1}^{SS} | h^m | \Phi_{vj,1}^{SS} \rangle \\
&= \frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) | \Phi_j^{\bar{v}} \rangle = -\frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_j^{\bar{v}} \rangle \\
&= -\frac{1}{\sqrt{2}} (\delta_{ij} h_{uv}^m - \delta_{uv} h_{ji}^m) \tag{245}
\end{aligned}$$

$$\begin{aligned}
53. \quad & \langle \Phi_{ui,1}^{SS} | h^m | \Phi_{bv,1}^{SS} \rangle \\
&= \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{T}_{pq}^\dagger(1, 0) | \Phi_u^a \rangle = 0 \tag{246}
\end{aligned}$$

$$\begin{aligned}
54. \quad & \langle \Phi_{ui,1}^{SS} | h^m | \Phi_{bj,2}^{SS} \rangle \\
&= \sqrt{\frac{S}{2(1+S)}} \sum_{pq} h_{pq}^m \left[-\overbrace{\langle \Phi_i^{\bar{u}} | \hat{T}_{pq}^\dagger(1, 0) | \Phi_j^{\bar{b}} \rangle}^{=0} + \langle \Phi_i^{\bar{u}} | \hat{T}_{pq}^\dagger(1, 0) | \Phi_j^{\bar{b}} \rangle - \frac{1}{S} \sum_v \langle \Phi_i^{\bar{u}} | \hat{T}_{pq}^\dagger(1, 0) | \Phi_{jv}^{\bar{b}\bar{v}} \rangle \right] \\
&= \sqrt{\frac{S}{2(1+S)}} \left[-\frac{1}{\sqrt{2}} \delta_{ij} h_{ub}^m + \frac{1}{S\sqrt{2}} \sum_{pq,v} h_{pq}^m \langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{u}_\beta) (\hat{p}_\alpha^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{v}_\beta^\dagger \hat{v}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle \right] \\
&= \frac{1}{\sqrt{2}} \sqrt{\frac{S}{2(1+S)}} \left[-\delta_{ij} h_{ub}^m + \frac{1}{S} \delta_{ij} h_{ub}^m \right] = \frac{1-S}{2\sqrt{S(S+1)}} \delta_{ij} h_{ub}^m \tag{247}
\end{aligned}$$

$$\begin{aligned}
55. \quad & \langle \Phi_{ui,1}^{SS} | h^m | \Phi_{bj}^{(S+1)(S+1)} \rangle \\
&= -\sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | \hat{T}_{pq}^\dagger(1, -1) | \Phi_j^{\bar{b}} \rangle
\end{aligned}$$

$$= - \sum_{pq} h_{pq}^m \langle \Phi_i^{\bar{u}} | (\hat{u}_\beta^\dagger \hat{u}_\beta) (\hat{p}_\beta^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{j}_\beta) | \Phi_j^b \rangle = -\delta_{ij} h_{ub}^m \quad (248)$$

56. $\langle \Phi_{au,1}^{SS} | h^m | \Phi_{bv,1}^{SS} \rangle$

$$= \sum_{pq} h_{pq}^m \langle \Phi_u^a | \hat{T}_{pq}^\dagger(1,0) | \Phi_v^b \rangle = \frac{1}{\sqrt{2}} (\delta_{uv} h_{ab}^m - \delta_{ab} h_{vu}^m) \quad (249)$$

57. $\langle \Phi_{au,1}^{SS} | h^m | \Phi_{bj,2}^{SS} \rangle$

$$\begin{aligned} &= \sqrt{\frac{S}{2(S+1)}} \sum_{pq} h_{pq}^m \left[\langle \Phi_u^a | -\hat{T}_{pq}^\dagger(1,0) | \Phi_j^b \rangle - \frac{1}{S} \sum_u \langle \Phi_u^a | -\hat{T}_{pq}^\dagger(1,0) | \Phi_{\bar{j}v}^{b\bar{v}} \rangle \right] \\ &= \sqrt{\frac{S}{S+1}} \frac{1}{2} \left[\delta_{ab} h_{ju}^m - \frac{1}{S} \delta_{ab} h_{ju}^m \right] = \frac{S-1}{2\sqrt{S(S+1)}} \delta_{ab} h_{ju}^m \end{aligned} \quad (250)$$

58. $\langle \Phi_{au,1}^{SS} | h^m | \Phi_{bj}^{(S+1)(S+1)} \rangle$

$$\begin{aligned} &= - \sum_{pq} h_{pq}^m \langle \Phi_u^a | \hat{T}_{pq}^\dagger(1,-1) | \Phi_j^b \rangle \\ &= - \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{u}_\alpha^\dagger \hat{a}_\alpha) (\hat{p}_\beta^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{a}_j) | \Phi_0^{SS} \rangle = \delta_{ab} h_{ju}^m \end{aligned} \quad (251)$$

59. $\langle \Phi_{ai,2}^{SS} | h^m | \Phi_{bj,2}^{SS} \rangle$

$$\begin{aligned} &= \frac{S}{2(1+S)} \left[\begin{aligned} &\langle \Phi_i^a | h^m | \Phi_j^b \rangle + \langle \Phi_i^{\bar{a}} | h^m | \Phi_j^{\bar{b}} \rangle \quad \rightarrow \quad \delta_{ij} h_{ab}^m - \delta_{ij} h_{ab}^m = 0 \\ &+ \langle \Phi_i^{\bar{a}} | h^m | \Phi_j^b \rangle + \langle \Phi_i^a | h^m | \Phi_j^{\bar{b}} \rangle \quad \rightarrow \quad 0 + 0 = 0 \\ &+ \frac{1}{S} \sum_v \langle \Phi_i^a | h^m | \Phi_{\bar{j}v}^{b\bar{v}} \rangle \quad \rightarrow \quad 0 \\ &- \frac{1}{S} \sum_v \langle \Phi_i^{\bar{a}} | h^m | \Phi_{\bar{j}v}^{b\bar{v}} \rangle \quad \rightarrow \quad 0 \\ &+ \frac{1}{S} \sum_u \langle \Phi_{iu}^{a\bar{u}} | h^m | \Phi_j^b \rangle \quad \rightarrow \quad 0 \end{aligned} \right] \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{S} \sum_u \langle \Phi_{iu}^{a\bar{u}} | h^m | \Phi_j^{\bar{b}} \rangle \rightarrow 0 \\
& + \frac{1}{S^2} \sum_{uv} \langle \Phi_{iu}^{a\bar{u}} | h^m | \Phi_{jv}^{b\bar{v}} \rangle \Big] = \frac{1}{2S(1+S)} \sum_{uv} \langle \Phi_{iu}^{a\bar{u}} | h^m | \Phi_{jv}^{b\bar{v}} \rangle \quad (252)
\end{aligned}$$

then the only one term(last term)

$$\frac{1}{2\sqrt{2}S(1+S)} \sum_{uv} \langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{u}_\beta \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{b}_\alpha^\dagger \hat{v}_\beta^\dagger \hat{v}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle \quad (253)$$

there are two situations to contract the above matrix element:

- pq contracted with u or v : this kind of terms should equal to zero, because:

$$\sum_{pq,uv} h_{pq}^m \langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{u}_\beta \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{v}_\beta^\dagger \hat{v}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle = -\delta_{ab} \delta_{ij} \overbrace{\sum_{uv} h_{uv}^m \delta_{uv}}^{=0} \quad (254)$$

the last equality is due to that the orbitals are real.

- pq contracted without u or v :

$$\begin{aligned}
(253) &= \frac{1}{2\sqrt{2}S(1+S)} \sum_{pq,uv} h_{pq}^m \Big[\langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{u}_\beta \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{v}_\beta^\dagger \hat{v}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle \\
&\quad - \langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{u}_\alpha^\dagger \hat{u}_\beta \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha) (\hat{b}_\alpha^\dagger \hat{v}_\beta^\dagger \hat{v}_\alpha \hat{j}_\beta) | \Phi_0^{SS} \rangle \Big] \\
&= \frac{1}{\sqrt{2}(1+S)} (\delta_{ij} h_{ab}^m + \delta_{ab} h_{ji}^m) \quad (255)
\end{aligned}$$

where we use the relation $\sum_{uv=1}^{2S} \delta_{uv} = 2S$.

60. $\langle \Phi_{ai,2}^{SS} | h^m | \Phi_{bj}^{(S+1)(S+1)} \rangle$

$$\begin{aligned}
&= \sqrt{\frac{S}{2(S+1)}} \Big[\langle \Phi_i^a | h^m | \Phi_j^b \rangle - \langle \Phi_i^{\bar{a}} | h^m | \Phi_j^{\bar{b}} \rangle + \frac{1}{S} \sum_u \langle \Phi_{iu}^{a\bar{u}} | h^m | \Phi_j^b \rangle \Big] \\
&= \sqrt{\frac{S}{2(S+1)}} \Big[-\delta_{ab} h_{ji}^m - \delta_{ij} h_{ab}^m + \delta_{ij} \delta_{ab} \sum_u h_{uu}^m \Big]
\end{aligned}$$

$$= -\sqrt{\frac{S}{2(S+1)}}(\delta_{ab}h_{ji}^m + \delta_{ij}h_{ab}^m) \quad (256)$$

$$\begin{aligned}
61. \quad & \langle \Phi_{ai}^{(S+1)(S+1)} | h^m | \Phi_{bj}^{(S+1)(S+1)} \rangle \\
&= \sum_{pq} h_{pq}^m \langle \Phi_i^a | \hat{T}(1, 0) | \Phi_j^b \rangle = \frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_i^a | \hat{T}(1, 0) | \Phi_j^b \rangle \\
&= \frac{1}{\sqrt{2}} \sum_{pq} h_{pq}^m \langle \Phi_0^{SS} | (\hat{i}_\beta^\dagger \hat{a}_\alpha) (\hat{p}_\alpha^\dagger \hat{q}_\alpha - \hat{p}_\beta^\dagger \hat{q}_\beta) (\hat{b}_\alpha^\dagger \hat{j}_\beta) | \Phi_0^{SS} \rangle \\
&= \frac{1}{\sqrt{2}} (\delta_{ij} h_{ab}^m + \delta_{ab} h_{ji}^m) \quad (257)
\end{aligned}$$

D. Summary of matrix elements

a. Property of $\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle$ The matrix $\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle$ have this property (use Eq. (14) and Eq. (18)):

$$\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle^* = \left(\sum_{pq} h_{pq}^m \langle \Phi_K^{SS} | \hat{T}_{pq}^\dagger(1, S - S') | \Phi_L^{S'S'} \rangle \right)^\dagger \quad (258)$$

$$= \sum_{pq} (h_{pq}^m)^* \left(\langle \Phi_K^{SS} | \hat{T}_{pq}^\dagger(1, S - S') | \Phi_L^{S'S'} \rangle \right)^\dagger \quad (259)$$

$$= \sum_{pq} \bar{h}_{pq}^m \langle \Phi_L^{S'S'} | (-1)^{S-S'} \hat{T}_{qp}^\dagger(1, S' - S) | \Phi_K^{SS} \rangle \quad (260)$$

$$= (-1)^{S-S'} \sum_{pq} (-\bar{h}_{qp}^m) \langle \Phi_L^{S'S'} | \hat{T}_{qp}^\dagger(1, S' - S) | \Phi_K^{SS} \rangle \quad (261)$$

$$= -(-1)^{S-S'} \sum_{pq} \bar{h}_{pq}^m \langle \Phi_L^{S'S'} | \hat{T}_{pq}^\dagger(1, S' - S) | \Phi_K^{SS} \rangle \quad (262)$$

where $\bar{h}_{pq}^m = (h_{pq}^m)^*$, which is means

$$\begin{aligned}
\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle &= \left(-(-1)^{S-S'} \sum_{pq} \bar{h}_{pq}^m \langle \Phi_L^{S'S'} | \hat{T}_{pq}^\dagger(1, S' - S) | \Phi_K^{SS} \rangle \right)^* \\
&= -(-1)^{S-S'} \langle \Phi_L^{S'S'} | \hat{h}^m | \Phi_K^{SS} \rangle \quad (263)
\end{aligned}$$

When $S' = S$ is satisfied, then the equation above can be simplified as

$$\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{SS} \rangle = -\langle \Phi_L^{SS} | \hat{h}^m | \Phi_K^{SS} \rangle \quad (264)$$

and

$$\langle \Phi_K^{(S\pm 1)(S\pm 1)} | \hat{h}^m | \Phi_L^{SS} \rangle = \langle \Phi_L^{SS} | \hat{h}^m | \Phi_K^{(S\pm 1)(S\pm 1)} \rangle \quad (265)$$

with the help of the property above, only the upper triangular part of this matrix $\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle$ is need.

b. Property of $\langle \Psi_I | \hat{h}^{\text{SO}} | \Psi_J \rangle$

$$\langle \Psi_J | \hat{h}^{\text{SO}} | \Psi_I \rangle = \langle \Psi_I | \hat{h}^{\text{SO}} | \Psi_J \rangle^* \quad (266)$$

This indicates that when constructing the matrix $\langle \Psi_I | \hat{h}^{\text{SO}} | \Psi_J \rangle$, we only need to fill the upper triangular part.

TABLE I. Matrix elements $\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle$, where $S' = S - 1$ CV(1) interaction and CO(1) interaction

No.	Bra	Ket	Matrix element
1.	(S-1)(CV1)	(S-1)(CV1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj}^{(S-1)(S-1)} \rangle = \frac{1-S}{S\sqrt{2}} (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab})$
2.	(S-1)(CV1)	(S-1)(CO1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{(S-1)(S-1)} \rangle = \frac{1-S}{2S} \sqrt{\frac{2S+1}{S}} h_{at}^m \delta_{ij}$
3.	(S-1)(CV1)	(S-1)(OV1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{(S-1)(S-1)} \rangle = \frac{1-S}{2S} \sqrt{\frac{2S+1}{S}} h_{ti}^m \delta_{ab}$
4.	(S-1)(CV1)	(S-1)(O1O2)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{tw}^{(S-1)(S-1)} \rangle = 0$
5.	(S-1)(CV1)	(S-1)(O1O1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_t^{(S-1)(S-1)} \rangle = 0$
6.	(S-1)(CV1)	GS	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_0^{SS} \rangle = \sqrt{\frac{2S-1}{2S+1}} h_{ai}^m$
7.	(S-1)(CV1)	S(CV0)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = \frac{1}{\sqrt{2}} \sqrt{\frac{2S-1}{2S+1}} (h_{ab}^m \delta_{ij} - h_{ji}^m \delta_{ab})$
8.	(S-1)(CV1)	S(CO0)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{SS} \rangle = -\frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} h_{at}^m \delta_{ij}$
9.	(S-1)(CV1)	S(OV0)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{SS} \rangle = \frac{1}{2S} \sqrt{\frac{2S-1}{2S+1}} h_{ti}^m \delta_{ab}$
10.	(S-1)(CV1)	S(CV1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = -\sqrt{\frac{(1+S)(2S-1)}{2S(2S+1)}} (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab})$
11.	(S-1)(CO1)	(S-1)(CO1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{(S-1)(S-1)} \rangle = -\frac{S-1}{S\sqrt{2}} (h_{ji}^m \delta_{ut} + \frac{2S+1}{2S-1} h_{ut}^m \delta_{ij})$
12.	(S-1)(CO1)	(S-1)(OV1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{(S-1)(S-1)} \rangle = 0$
13.	(S-1)(CO1)	(S-1)(O1O2)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{tw}^{(S-1)(S-1)} \rangle = -\frac{S-1}{\sqrt{S(2S-1)}} h_{wt}^m \delta_{ut}$
14.	(S-1)(CO1)	(S-1)(O1O1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_t^{(S-1)(S-1)} \rangle = -\frac{1}{2\sqrt{S(2S-1)}} h_{ui}^m \left(\frac{1-S}{S} + 2(S-1) \delta_{ut} \right)$
15.	(S-1)(CO1)	GS	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_0^{SS} \rangle = \sqrt{\frac{2S-1}{2S}} h_{ui}^m$
16.	(S-1)(CO1)	S(CV0)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = \frac{1}{2} \sqrt{\frac{2S-1}{S}} h_{ub}^m \delta_{ij}$
17.	(S-1)(CO1)	S(CO0)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{SS} \rangle = -\sqrt{\frac{2S-1}{2S}} \left(h_{ji}^m \delta_{ut} + \frac{1}{2S-1} h_{ut}^m \delta_{ij} \right)$
18.	(S-1)(CO1)	S(OV0)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{SS} \rangle = 0$
19.	(S-1)(CO1)	S(CV1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = -\frac{\sqrt{(S+1)(2S-1)}}{2S} h_{ub}^m \delta_{ij}$

TABLE II. Matrix elements $\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle$, where $S' = S - 1$ OV(1), O₁O₂ and O₁O₁ interaction

No.	Bra	Ket	Matrix element
20.	(S-1)(OV1)	(S-1)(OV1)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{(S-1)(S-1)} \rangle = -\frac{S-1}{S\sqrt{2}}(h_{ab}^m \delta_{ut} + \frac{2S+1}{2S-1} h_{tu}^m \delta_{ab})$
21.	(S-1)(OV1)	(S-1)(O1O2)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_{tw}^{(S-1)(S-1)} \rangle = \frac{1-S}{\sqrt{S(2S-1)}} h_{at}^m \delta_{uw}$
22.	(S-1)(OV1)	(S-1)(O1O1)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_t^{(S-1)(S-1)} \rangle = -\frac{1}{2\sqrt{S(2S-1)}} h_{au}^m \left(\frac{1-S}{S} + 2(S-1) \delta_{ut} \right)$
23.	(S-1)(OV1)	GS	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_0^{SS} \rangle = \sqrt{\frac{2S-1}{2S}} h_{au}^m$
24.	(S-1)(OV1)	S(CV0)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = -\frac{1}{2} \sqrt{\frac{2S-1}{S}} h_{ju}^m \delta_{ab}$
25.	(S-1)(OV1)	S(CO0)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{SS} \rangle = 0$
26.	(S-1)(OV1)	S(OV0)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{SS} \rangle = \sqrt{\frac{2S-1}{2S}} \left(h_{ab}^m \delta_{ut} + \frac{1}{2S-1} h_{tu}^m \delta_{ab} \right)$
27.	(S-1)(OV1)	S(CV1)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = -\frac{\sqrt{(S+1)(2S-1)}}{2S} h_{ju}^m \delta_{ab}$
28.	(S-1)(O1O2)	(S-1)(O1O2)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_{tw}^{(S-1)(S-1)} \rangle = -\frac{1}{\sqrt{2}}(h_{uv}^m \delta_{ut} + h_{ut}^m \delta_{vw})$
29.	(S-1)(O1O2)	(S-1)(O1O1)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_t^{(S-1)(S-1)} \rangle = -\frac{1}{\sqrt{2}}(h_{uv}^m \delta_{ut} + h_{uv}^m (\delta_{vt} - \frac{1}{S}))$
30.	(S-1)(O1O2)	GS	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_0^{SS} \rangle = h_{uv}^m$
31.	(S-1)(O1O2)	S(CV0)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = 0$
32.	(S-1)(O1O2)	S(CO0)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{SS} \rangle = -h_{jv}^m \delta_{tu}$
33.	(S-1)(O1O2)	S(OV0)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{SS} \rangle = h_{ub}^m \delta_{vt}$
34.	(S-1)(O1O2)	S(CV1)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = 0$
35.	(S-1)(O1O1)	(S-1)(O1O1)	$\langle \Phi_u^{(S-1)(S-1)} \hat{h}^m \Phi_t^{(S-1)(S-1)} \rangle = 0$
36.	(S-1)(O1O1)	GS	$\langle \Phi_u^{(S-1)(S-1)} \hat{h}^m \Phi_0^{SS} \rangle = 0$
37.	(S-1)(O1O1)	S(CV0)	$\langle \Phi_u^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = 0$
38.	(S-1)(O1O1)	S(CO0)	$\langle \Phi_u^{(S-1)(S-1)} \hat{h}^m \Phi_{tj}^{SS} \rangle = -h_{jt}^m (\delta_{ut} - \frac{1}{2S})$
39.	(S-1)(O1O1)	S(OV0)	$\langle \Phi_u^{(S-1)(S-1)} \hat{h}^m \Phi_{bt}^{SS} \rangle = h_{tb}^m (\delta_{ut} - \frac{1}{2S})$
40.	(S-1)(O1O1)	S(CV1)	$\langle \Phi_u^{(S-1)(S-1)} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = 0$

TABLE III. Matrix elements $\langle \Phi_K^{SS} | \hat{h}^m | \Phi_L^{S'S'} \rangle$ with $S' = S$ or $S' = S+1$ state interaction

No.	Bra	Ket	Matrix element
41.	GS	GS	$\langle \Phi_0^{SS} \hat{h}^m \Phi_0^{SS} \rangle = 0$
42.	GS	S(CV0)	$\langle \Phi_0^{SS} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = 0$
43.	GS	S(CO0)	$\langle \Phi_0^{SS} \hat{h}^m \Phi_{vj,1}^{SS} \rangle = -\frac{1}{\sqrt{2}} h_{jv}^m$
44.	GS	S(OV0)	$\langle \Phi_0^{SS} \hat{h}^m \Phi_{bv,1}^{SS} \rangle = \frac{1}{\sqrt{2}} h_{vb}^m$
45.	GS	S(CV1)	$\langle \Phi_0^{SS} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = -\sqrt{\frac{S}{(1+S)}} h_{jb}^m$
46.	GS	(S+1)(CV1)	$\langle \Phi_0^{SS} \hat{h}^m \Phi_{bj}^{(S+1)(S+1)} \rangle = -h_{jb}^m$
47.	S(CV0)	S(CV0)	$\langle \Phi_{ai,1}^{SS} \hat{h}^m \Phi_{bj,1}^{SS} \rangle = 0$
48.	S(CV0)	S(CO0)	$\langle \Phi_{ai,1}^{SS} \hat{h}^m \Phi_{vj,1}^{SS} \rangle = -\frac{1}{2} h_{av}^m \delta_{ij}$
49.	S(CV0)	S(OV0)	$\langle \Phi_{ai,1}^{SS} \hat{h}^m \Phi_{bv,1}^{SS} \rangle = -\frac{1}{2} h_{vi}^m \delta_{ab}$
50.	S(CV0)	S(CV1)	$\langle \Phi_{ai,1}^{SS} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = -\sqrt{\frac{S}{2(1+S)}} (h_{ab}^m \delta_{ij} - h_{ji}^m \delta_{ab})$
51.	S(CV0)	(S+1)(CV1)	$\langle \Phi_{ai,1}^{SS} \hat{h}^m \Phi_{bj}^{(S+1)(S+1)} \rangle = \frac{1}{\sqrt{2}} (h_{ji}^m \delta_{ab} - h_{ab}^m \delta_{ij})$
52.	S(CO0)	S(CO0)	$\langle \Phi_{ui}^{SS} \hat{h}^m \Phi_{vj,1}^{SS} \rangle = -\frac{1}{\sqrt{2}} (h_{uv}^m \delta_{ij} - h_{ji}^m \delta_{uv})$
53.	S(CO0)	S(OV0)	$\langle \Phi_{ui}^{SS} \hat{h}^m \Phi_{bv,1}^{SS} \rangle = 0$
54.	S(CO0)	S(CV1)	$\langle \Phi_{ui}^{SS} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = \frac{1-S}{2\sqrt{S(S+1)}} h_{ub}^m \delta_{ij}$
55.	S(CO0)	(S+1)(CV1)	$\langle \Phi_{ui}^{SS} \hat{h}^m \Phi_{bj}^{(S+1)(S+1)} \rangle = -h_{ub}^m \delta_{ij}$
56.	S(OV0)	S(OV0)	$\langle \Phi_{au}^{SS} \hat{h}^m \Phi_{bv,1}^{SS} \rangle = \frac{1}{\sqrt{2}} (h_{ab}^m \delta_{uv} - h_{vu}^m \delta_{ab})$
57.	S(OV0)	S(CV1)	$\langle \Phi_{au}^{SS} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = \frac{S-1}{2\sqrt{S(S+1)}} h_{ju}^m \delta_{ab}$
58.	S(OV0)	(S+1)(CV1)	$\langle \Phi_{au}^{SS} \hat{h}^m \Phi_{bj}^{(S+1)(S+1)} \rangle = h_{ju}^m \delta_{ab}$
59.	S(CV1)	S(CV1)	$\langle \Phi_{ai,2}^{SS} \hat{h}^m \Phi_{bj,2}^{SS} \rangle = \frac{1}{\sqrt{2(1+S)}} (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab})$
60.	S(CV1)	(S+1)(CV1)	$\langle \Phi_{ai,2}^{SS} \hat{h}^m \Phi_{bj}^{(S+1)(S+1)} \rangle = -\sqrt{\frac{S}{2(S+1)}} (h_{ji}^m \delta_{ab} + h_{ab}^m \delta_{ij})$
61.	(S+1)(CV1)	(S+1)(CV1)	$\langle \Phi_{ai}^{(S+1)(S+1)} \hat{h}^m \Phi_{bj}^{(S+1)(S+1)} \rangle = \frac{1}{\sqrt{2}} (h_{ab}^m \delta_{ij} + h_{ji}^m \delta_{ab})$

IV. CALCULATION OF OSCILLATOR STRENGTH

The oscillator strength between state $|\Psi_I^{\text{SO}}\rangle$ and $|\Phi_J^{\text{SO}}\rangle$ (where $I = 0$ or $J = 0$ means that the corresponding state $|\Psi_0\rangle = |\Phi_0\rangle$ is the reference, and where the excited states is under consideration of SOC) reads

$$f_{IJ} = \frac{2}{3}(E_J^{\text{SO}} - E_I^{\text{SO}}) \sum_{i=1}^3 |\mu_{IJ}^i|^2 \quad (267)$$

where μ_{IJ}^i is the dipole moment between those two states.

$$\mu_{IJ}^i = \sum_{pq} h_{pq}^i \langle \Psi_I^{\text{SO}} | \hat{E}_{pq} | \Psi_J^{\text{SO}} \rangle, \quad i = x, y, z \quad (268)$$

here the electronic integral

$$h_{pq}^i = \int_{\mathbb{R}^3} \phi_p^*(\vec{x}) r_i \phi_q(\vec{x}) d\vec{x}, \quad h_{pq}^i = h_{qp}^i \quad (269)$$

the question is how to calculate $\gamma_{pq}(I, J) := \langle \Psi_I^{\text{SO}} | \hat{E}_{pq} | \Psi_J^{\text{SO}} \rangle$, i.e. TDM1. Note that the state $|\Psi_I^{\text{SO}}\rangle$ is not necessarily a eigenstate of $\hat{S}^2$ operator, we can do calculation in the spin-adapted basis. The matrix elements can be expanded on the unperturbed excited state

$$\langle \Psi_I^{\text{SO}} | \hat{E}_{pq} | \Psi_J^{\text{SO}} \rangle = \sum_{LK} X'_{LI} \langle \Phi_L | \hat{E}_{pq} | \Phi_K \rangle X'_{KJ} \quad (270)$$

and

$$|\Psi_I^{\text{SO}}\rangle = \sum_K \langle \Psi_K | \Psi_I^{\text{SO}} \rangle |\Psi_K\rangle = \overbrace{\sum_{KL} X'_{LK} \langle \Psi_K | \Psi_I^{\text{SO}} \rangle}^{X'_{LI}} |\Phi_L\rangle \quad (271)$$

here $\langle \Psi_K | \Psi_I^{\text{SO}} \rangle$ can obtained by SI calculation.

This issue is not as difficult as that in the calculation of matrix elements of $\hat{h}^m$, thanks to the selection rule of spin (Wigner–Eckart theorem)

$$\langle \Phi_L^{S'M'} | \hat{E}_{pq} | \Phi_K^{SM} \rangle = \langle S'M'00 | SM \rangle \langle \Phi_L^S | \hat{E}_{pq} | \Phi_K^{S'} \rangle \quad (272)$$

because of

$$\langle S'M'00 | SM \rangle = \delta_{SS'} \delta_{MM'}, \quad (273)$$

then

$$\langle \Phi_L^{SM} | \hat{E}_{pq} | \Phi_K^{SM} \rangle = \langle \Phi_L^S | \hat{E}_{pq} | \Phi_K^S \rangle = \langle \Phi_L^{SS} | \hat{E}_{pq} | \Phi_K^{SS} \rangle \quad (274)$$

we only need to consider those four kind of matrix elements

$$\langle \Phi_L^{(S-1)(S-1)} | \hat{E}_{pq} | \Phi_K^{(S-1)(S-1)} \rangle \quad (275)$$

$$\langle \Phi_L^{SS} | \hat{E}_{pq} | \Phi_K^{SS} \rangle, \quad L, K = 0, 1, \dots \quad (276)$$

$$\langle \Phi_L^{(S+1)(S+1)} | \hat{E}_{pq} | \Phi_K^{(S+1)(S+1)} \rangle \quad (277)$$

and then

$$\langle \Psi_I^{\text{SO}} | \hat{O} | \Psi_J^{\text{SO}} \rangle = \sum_{KL} \langle \Psi_I^{\text{SO}} | \Psi_K \rangle \langle \Psi_K | \hat{O} | \Psi_L \rangle \langle \Psi_L | \Psi_J^{\text{SO}} \rangle \quad (278)$$

$$= \sum_{KL} \langle \Psi_K | \hat{O} | \Psi_L \rangle \langle \Psi_K | \Psi_I^{\text{SO}} \rangle^\dagger \langle \Psi_L | \Psi_J^{\text{SO}} \rangle \quad (279)$$

here

$$\langle \Psi_K | \hat{O} | \Psi_L \rangle = \sum_{K'L'} X_{L',L}^\dagger \sum_{pq} o_{pq} \langle \Phi_{K'} | \hat{E}_{pq} | \Phi_{L'} \rangle X_{K',K} \quad (280)$$

The remaining task is to derive the explicit form of the matrix elements $\langle \Phi_I | \hat{E}_{pq} | \Phi_J \rangle$. We can derive the formulation of $\sum_{pq} o_{pq} \langle \Phi_I | \hat{E}_{pq} | \Phi_J \rangle$, then

$$\gamma_{pq}(I, J) = \frac{\partial}{\partial o_{pq}} \langle \Phi_I | \hat{O} | \Phi_J \rangle = \frac{\partial}{\partial o_{pq}} \sum_{rs} o_{rs} \langle \Phi_I | \hat{E}_{rs} | \Phi_J \rangle \quad (281)$$

A. $|S' = S - 1\rangle$ block

1. $\langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_{bj}^{(S-1)(S-1)} \rangle$

$$\sum_{pq} o_{pq} \langle \Phi_{ai}^{\bar{a}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha + \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{bj}^{\bar{b}} \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab} \quad (282)$$

$$\sum_{pq} \sum_{xy} o_{pq} \langle \Phi_{ix}^{a\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha + \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jy}^{b\bar{y}} \rangle = (2S)(o_{ab} \delta_{ij} - o_{ji} \delta_{ab}) \quad (283)$$

$$\sum_{pq} \sum_{xy} o_{pq} \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha + \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jy}^{\bar{b}\bar{y}} \rangle = (2S)(o_{ab} \delta_{ij} - o_{ji} \delta_{ab}) \quad (284)$$

$$\sum_{pq} \sum_{\substack{x \neq y \\ z \neq w}} o_{pq} \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{p}_\alpha^\dagger \hat{q}_\alpha + \hat{p}_\beta^\dagger \hat{q}_\beta | \Phi_{jzw}^{b\bar{z}\bar{w}} \rangle = 2(2S)(2S-1)(o_{ab}\delta_{ij} - o_{ji}\delta_{ab}) \quad (285)$$

Finally, this matrix element reads

$$\begin{aligned} \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_{bj}^{(S-1)(S-1)} \rangle &= \left(\frac{2S-1}{2S+1} + \frac{1}{S} \frac{2S-1}{2S+1} + \frac{1}{S(2S+1)} \right) (o_{ab}\delta_{ij} - o_{ji}\delta_{ab}) \\ &= o_{ab}\delta_{ij} - o_{ji}\delta_{ab} \end{aligned} \quad (286)$$

$$2. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_{tj}^{(S-1)(S-1)} \rangle$$

$$\langle \Phi_i^{\bar{a}} | \hat{O} | \Phi_j^{\bar{t}} \rangle = o_{at}\delta_{ij} \quad (287)$$

$$\sum_{pqx} \langle \Phi_{ix}^{a\bar{x}} | \hat{O} | \Phi_j^{\bar{t}} \rangle = -o_{at}\delta_{ij} \quad (288)$$

$$\sum_{pqx} \sum_{y \neq t} \langle \Phi_{ix}^{a\bar{x}} | \hat{O} | \Phi_{jy}^{\bar{t}\bar{y}} \rangle = (2S-1)o_{at}\delta_{ij} \quad (289)$$

$$\sum_{pq} \sum_{\substack{z \neq t \\ x \neq y}} \langle \Phi_{ixy}^{a\bar{x}\bar{y}} | \hat{O} | \Phi_{jz}^{\bar{t}\bar{z}} \rangle = -2(2S-1)o_{at}\delta_{ij} \quad (290)$$

Then the final result becomes

$$\begin{aligned} \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_{tj}^{(S-1)(S-1)} \rangle &= \frac{1}{\sqrt{2S(2S+1)}} \left(2S-1 + \frac{2S-1}{2S} + \frac{2S-1}{2S} + \frac{1}{S} \right) o_{at}\delta_{ij} \\ &= \sqrt{\frac{2S+1}{2S}} o_{at}\delta_{ij} \end{aligned} \quad (291)$$

$$3. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_{bt}^{(S-1)(S-1)} \rangle$$

$$= -\sqrt{\frac{2S+1}{2S}} o_{ti}\delta_{ab} \quad (292)$$

$$4. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_{tw}^{(S-1)(S-1)} \rangle = 0$$

$$5. \langle \Phi_{ai}^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle = 0$$

$$\langle \Phi_i^{\bar{a}} | \hat{O} | \Phi_t^{\bar{t}} \rangle = 0 \quad (293)$$

$$\sum_x \langle \Phi_{ix}^{a\bar{x}} | \hat{O} | \Phi_t^{\bar{t}} \rangle = o_{ai} \quad (294)$$

$$\sum_x \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{O} | \Phi_t^{\bar{t}} \rangle = o_{ai} \quad (295)$$

$$\sum_y \langle \Phi_i^{\bar{a}} | \hat{O} | \Phi_y^{\bar{y}} \rangle = 0 \quad (296)$$

$$\sum_{xy} \langle \Phi_{ix}^{a\bar{x}} | \hat{O} | \Phi_y^{\bar{y}} \rangle = (2S)o_{ai} \quad (297)$$

$$\sum_{xy} \langle \Phi_{ix}^{\bar{a}\bar{x}} | \hat{O} | \Phi_y^{\bar{y}} \rangle = (2S)o_{ai} \quad (298)$$

$$6. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{tj}^{(S-1)(S-1)} \rangle$$

$$\langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_j^{\bar{t}} \rangle = o_{ut}\delta_{ij} - o_{ji}\delta_{ut} \quad (299)$$

$$\sum_{y \neq t} \langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_{jy}^{\bar{t}\bar{y}} \rangle = 0 \quad (300)$$

$$\sum_{x \neq u} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_j^{\bar{t}} \rangle = 0 \quad (301)$$

$$\sum_{\substack{x \neq u \\ y \neq t}} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_{jy}^{\bar{t}\bar{y}} \rangle = (2S-1)(o_{ut}\delta_{ij} - o_{ji}\delta_{ut}) \quad (302)$$

then

$$\begin{aligned} \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{tj}^{(S-1)(S-1)} \rangle &= \left(\frac{2S-1}{2S} + \frac{1}{2S} \right) (o_{ut}\delta_{ij} - o_{ji}\delta_{ut}) \\ &= o_{ut}\delta_{ij} - o_{ji}\delta_{ut} \end{aligned} \quad (303)$$

$$7. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{bt}^{(S-1)(S-1)} \rangle = 0$$

$$\langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_t^{\bar{b}} \rangle = 0 \quad (304)$$

$$\sum_{y \neq t} \langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_{ty}^{b\bar{y}} \rangle = 0 \quad (305)$$

$$\sum_{x \neq u} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_t^{\bar{b}} \rangle = 0 \quad (306)$$

$$\sum_{\substack{x \neq u \\ y \neq t}} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_{ty}^{b\bar{y}} \rangle = 0 \quad (307)$$

(because i and b are both creation operator's index)

$$8. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{tw}^{(S-1)(S-1)} \rangle$$

$$\langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_w^{\bar{t}} \rangle = -o_{wi} \delta_{ut} \quad (308)$$

$$\sum_{x \neq u} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_w^{\bar{t}} \rangle = -o_{wi} \delta_{ut} \quad (309)$$

then

$$\langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{tw}^{(S-1)(S-1)} \rangle = -\sqrt{\frac{2S}{2S-1}} o_{wi} \delta_{ut} \quad (310)$$

$$9. \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle$$

$$\langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_t^{\bar{t}} \rangle = -o_{ti} \delta_{ut} = -o_{ui} \delta_{ut} \quad (311)$$

$$\sum_x \langle \Phi_i^{\bar{u}} | \hat{O} | \Phi_x^{\bar{x}} \rangle = -o_{ui} \quad (312)$$

$$\sum_{x \neq u} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_t^{\bar{t}} \rangle = \sum_{x \neq u} o_{ui} \delta_{xt} = o_{ui} - o_{ui} \delta_{ut} = (1 - \delta_{ut}) o_{ui} \quad (313)$$

$$\sum_y \sum_{x \neq u} \langle \Phi_{ix}^{\bar{u}\bar{x}} | \hat{O} | \Phi_y^{\bar{y}} \rangle = (2S) o_{ui} - o_{ui} = (2S - 1) o_{ui} \quad (314)$$

then

$$\begin{aligned} & \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle \\ &= -\sqrt{\frac{2S-1}{2S}} \left(\delta_{ut} - \frac{1}{2S} \right) o_{ui} + \frac{1}{\sqrt{2S(2S-1)}} \left(1 - \delta_{ut} - \frac{2S-1}{2S} \right) o_{ui} \\ &= \frac{1}{\sqrt{2S(2S-1)}} \left(\frac{2S-1}{2S} - (2S-1) \delta_{ut} + 1 - \delta_{ut} - 1 + \frac{1}{2S} \right) o_{ui} \\ &= \frac{1}{\sqrt{2S(2S-1)}} \left(1 - 2S \delta_{ut} \right) o_{ui} \end{aligned} \quad (315)$$

Here, we can divide

$$\langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{t,w}^{(S-1)(S-1)} \rangle$$

$$\begin{aligned}
&= (1 - \delta_{tw}) \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_{tw}^{(S-1)(S-1)} \rangle + \delta_{tw} \langle \Phi_{ui}^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle \\
&= - (1 - \delta_{tw}) \sqrt{\frac{2S}{2S-1}} o_{wi} \delta_{ut} + \frac{1}{\sqrt{2S(2S-1)}} (1 - 2S\delta_{ut}) \delta_{tw} o_{ui} \\
&= - \sqrt{\frac{2S}{2S-1}} o_{wi} \delta_{ut} + \frac{1}{\sqrt{2S(2S-1)}} \left((1 - 2S\delta_{ut}) \delta_{tw} o_{ui} + 2S\delta_{tw} o_{wi} \delta_{ut} \right) \\
&= - \sqrt{\frac{2S}{2S-1}} o_{wi} \delta_{ut} + \frac{1}{\sqrt{2S(2S-1)}} \delta_{tw} o_{sui} \tag{316}
\end{aligned}$$

here $|\Phi_{t,w}^{(S-1)(S-1)}\rangle$ means that the equality $t = w$ can be taken.

$$\begin{aligned}
10. \quad & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{O} | \Phi_{bt}^{(S-1)(S-1)} \rangle \\
&= o_{ab} \delta_{ut} - o_{tu} \delta_{ab} \tag{317}
\end{aligned}$$

$$\begin{aligned}
11. \quad & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{O} | \Phi_{tw}^{(S-1)(S-1)} \rangle \\
&= \sqrt{\frac{2S}{2S-1}} o_{at} \delta_{uw} \tag{318}
\end{aligned}$$

$$\begin{aligned}
12. \quad & \langle \Phi_{au}^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle \\
&= - \frac{1}{\sqrt{2S(2S-1)}} (1 - 2S\delta_{ut}) o_{at} \tag{319}
\end{aligned}$$

$$\begin{aligned}
13. \quad & \langle \Phi_{uv}^{(S-1)(S-1)} | \hat{O} | \Phi_{tw}^{(S-1)(S-1)} \rangle \\
&= \langle \Phi_v^{\bar{u}} | \hat{O} | \Phi_w^{\bar{t}} \rangle = o_{ut} \delta_{vw} - o_{wv} \delta_{ut} \tag{320}
\end{aligned}$$

$$\begin{aligned}
14. \quad & \langle \Phi_{uv}^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle = o_{ut} \delta_{vt} - o_{tv} \delta_{ut} \\
&\langle \Phi_v^{\bar{u}} | \hat{O} | \Phi_t^{\bar{t}} \rangle = o_{ut} \delta_{vt} - o_{tv} \delta_{ut} \tag{321}
\end{aligned}$$

$$\sum_x \langle \Phi_v^{\bar{u}} | \hat{O} | \Phi_x^{\bar{x}} \rangle = o_{uv} - o_{uv} = 0 \tag{322}$$

$$15. \quad \langle \Phi_u^{(S-1)(S-1)} | \hat{O} | \Phi_t^{(S-1)(S-1)} \rangle = 0$$

B. $|S' = S\rangle$ block

$$16. \langle \Phi_0^{SS} | \hat{O} | \Phi_0^{SS} \rangle = 0$$

$$17. \langle \Phi_0^{SS} | \hat{O} | \Phi_{bj,1}^{SS} \rangle = \sqrt{2} \langle \Phi_0 | \hat{O} | \Phi_j^b \rangle = \sqrt{2} o_{bj} \quad (323)$$

$$18. \langle \Phi_0^{SS} | \hat{O} | \Phi_{tj}^{SS} \rangle = \langle \Phi_0 | \hat{O} | \Phi_{\bar{j}}^{\bar{t}} \rangle = o_{jt} \quad (324)$$

$$19. \langle \Phi_0^{SS} | \hat{O} | \Phi_{bt}^{SS} \rangle = \langle \Phi_0 | \hat{O} | \Phi_t^b \rangle = o_{tb} \quad (325)$$

$$20. \langle \Phi_0^{SS} | \hat{O} | \Phi_{bj,2}^{SS} \rangle = 0$$

$$21. \langle \Phi_{ai,1}^{SS} | \hat{O} | \Phi_{bj,1}^{SS} \rangle = \langle \Phi_i^a | \hat{O} | \Phi_j^b \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab} \quad (326)$$

$$22. \langle \Phi_{ai,1}^{SS} | \hat{O} | \Phi_{tj}^{SS} \rangle = \frac{1}{\sqrt{2}} \langle \Phi_{\bar{i}}^{\bar{a}} | \hat{O} | \Phi_{\bar{j}}^{\bar{t}} \rangle = \frac{1}{\sqrt{2}} o_{at} \delta_{ij} \quad (327)$$

$$23. \langle \Phi_{ai,1}^{SS} | \hat{O} | \Phi_{bt}^{SS} \rangle = \frac{1}{\sqrt{2}} \langle \Phi_i^a | \hat{O} | \Phi_t^b \rangle = -\frac{1}{\sqrt{2}} o_{ti} \delta_{ab} \quad (328)$$

$$24. \langle \Phi_{ai,1}^{SS} | \hat{O} | \Phi_{bj,2}^{SS} \rangle = 0$$

$$25. \langle \Phi_{ui}^{SS} | \hat{O} | \Phi_{vj}^{SS} \rangle = \langle \Phi_{\bar{i}}^{\bar{u}} | \hat{O} | \Phi_{\bar{j}}^{\bar{v}} \rangle = o_{ut} \delta_{ij} - o_{ji} \delta_{ut} \quad (329)$$

$$26. \langle \Phi_{ui}^{SS} | \hat{O} | \Phi_{bv}^{SS} \rangle = \langle \Phi_{\bar{i}}^{\bar{u}} | \hat{O} | \Phi_t^b \rangle = 0 \quad (330)$$

$$27. \langle \Phi_{ui}^{SS} | \hat{O} | \Phi_{bj,2}^{SS} \rangle$$

$$\langle \Phi_{\bar{i}}^{\bar{u}} | \hat{O} | \Phi_j^b \rangle = 0 \quad (331)$$

$$\langle \Phi_{\bar{i}}^{\bar{u}} | \hat{O} | \Phi_j^{\bar{b}} \rangle = o_{ub} \delta_{ij} \quad (332)$$

$$\sum_x \langle \Phi_{\bar{i}}^{\bar{u}} | \hat{O} | \Phi_{jx}^{b\bar{x}} \rangle = -o_{ub} \delta_{ij} \quad (333)$$

then

$$\begin{aligned} \langle \Phi_{ui}^{SS} | \hat{O} | \Phi_{bj,2}^{SS} \rangle &= \sqrt{\frac{S}{2(1+S)}} \left(1 + \frac{1}{S}\right) o_{ub} \delta_{ij} \\ &= \sqrt{\frac{1+S}{2S}} o_{ub} \delta_{ij} \end{aligned} \quad (334)$$

$$28. \langle \Phi_{au}^{SS} | \hat{O} | \Phi_{bt}^{SS} \rangle$$

$$= \langle \Phi_u^a | \hat{O} | \Phi_t^b \rangle = o_{ab} \delta_{ut} - o_{tu} \delta_{ab} \quad (335)$$

$$29. \langle \Phi_{au}^{SS} | \hat{O} | \Phi_{bj,2}^{SS} \rangle$$

$$= \sqrt{\frac{1+S}{2S}} o_{ju} \delta_{ab} \quad (336)$$

$$30. \langle \Phi_{ai,2}^{SS} | \hat{O} | \Phi_{bj,2}^{SS} \rangle$$

$$\begin{aligned} &= \frac{2S}{2(1+S)} (o_{ab} \delta_{ij} - o_{ji} \delta_{ab}) + \sum_{xy} \langle \Phi_{ix}^{a\bar{x}} | \hat{O} | \Phi_{jy}^{b\bar{y}} \rangle \\ &= \frac{S}{2(1+S)} \left(\frac{2}{S} + 2 \right) (o_{ab} \delta_{ij} - o_{ji} \delta_{ab}) = o_{ab} \delta_{ij} - o_{ji} \delta_{ab} \end{aligned} \quad (337)$$

where

$$\sum_{xy} \langle \Phi_{ix}^{a\bar{x}} | \hat{O} | \Phi_{jy}^{b\bar{y}} \rangle \quad (338)$$

$$\begin{aligned} &= \sum_{xy} \left(-o_{yx} \delta_{xy} \delta_{ij} + o_{ab} \delta_{xy} \delta_{ij} - o_{ji} \delta_{xy} \delta_{ab} + o_{xy} \delta_{xy} \delta_{ij} \right) \\ &= \sum_{xy} \left(+o_{ab} \delta_{xy} \delta_{ij} - o_{ji} \delta_{xy} \delta_{ab} \right) = (2S)(o_{ab} \delta_{ij} - o_{ji} \delta_{ab}) \end{aligned} \quad (339)$$

C. $|S' = S + 1\rangle$ block

$$31. \langle \Phi_{ai}^{(S+1)(S+1)} | \hat{O} | \Phi_{bj}^{(S+1)(S+1)} \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab}$$

D. Summary of matrix elements

For the Hermitian operator $\hat{O} = \hat{O}^\dagger$ the matrix

$$\langle \Psi_I | \hat{O} | \Psi_J \rangle = \langle \Psi_J | \hat{O} | \Psi_I \rangle^\dagger \quad (340)$$

Thus, we only need to construct the upper-triangular part of the matrix, and symmetrize it afterward.

TABLE IV. Matrix elements $\langle \Phi_K^{S'S'} | \hat{O} | \Phi_L^{S'S'} \rangle$, where bra and ket are both $S' = S - 1$

No.	Bra	Ket	Matrix element
1.	(S-1)(CV1)	(S-1)(CV1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{O} \Phi_{bj}^{(S-1)(S-1)} \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab}$
2.	(S-1)(CV1)	(S-1)(CO1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{O} \Phi_{tj}^{(S-1)(S-1)} \rangle = \sqrt{\frac{2S+1}{2S}} o_{at} \delta_{ij}$
3.	(S-1)(CV1)	(S-1)(OV1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{O} \Phi_{bt}^{(S-1)(S-1)} \rangle = -\sqrt{\frac{2S+1}{2S}} o_{ti} \delta_{ab}$
4.	(S-1)(CV1)	(S-1)(O1O2)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{O} \Phi_{tw}^{(S-1)(S-1)} \rangle = 0$
5.	(S-1)(CV1)	(S-1)(O1O1)	$\langle \Phi_{ai}^{(S-1)(S-1)} \hat{O} \Phi_t^{(S-1)(S-1)} \rangle = 0$
6.	(S-1)(CO1)	(S-1)(CO1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{O} \Phi_{tj}^{(S-1)(S-1)} \rangle = o_{ut} \delta_{ij} - o_{ji} \delta_{ut}$
7.	(S-1)(CO1)	(S-1)(OV1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{O} \Phi_{bt}^{(S-1)(S-1)} \rangle = 0$
8.	(S-1)(CO1)	(S-1)(O1O2)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{O} \Phi_{tw}^{(S-1)(S-1)} \rangle = -\sqrt{\frac{2S}{2S-1}} o_{wi} \delta_{ut}$
9.	(S-1)(CO1)	(S-1)(O1O1)	$\langle \Phi_{ui}^{(S-1)(S-1)} \hat{O} \Phi_t^{(S-1)(S-1)} \rangle = \frac{1}{\sqrt{2S(2S-1)}} (1 - 2S \delta_{ut}) o_{ui}$
10.	(S-1)(OV1)	(S-1)(OV1)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{O} \Phi_{bt}^{(S-1)(S-1)} \rangle = o_{ab} \delta_{ut} - o_{tu} \delta_{ab}$
11.	(S-1)(OV1)	(S-1)(O1O2)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{O} \Phi_{tw}^{(S-1)(S-1)} \rangle = \sqrt{\frac{2S}{2S-1}} o_{at} \delta_{uw}$
12.	(S-1)(OV1)	(S-1)(O1O1)	$\langle \Phi_{au}^{(S-1)(S-1)} \hat{O} \Phi_t^{(S-1)(S-1)} \rangle = -\frac{1}{\sqrt{2S(2S-1)}} (1 - 2S \delta_{ut}) o_{at}$
13.	(S-1)(O1O2)	(S-1)(O1O2)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{O} \Phi_{tw}^{(S-1)(S-1)} \rangle = o_{ut} \delta_{vw} - o_{wv} \delta_{ut}$
14.	(S-1)(O1O2)	(S-1)(O1O1)	$\langle \Phi_{uv}^{(S-1)(S-1)} \hat{O} \Phi_t^{(S-1)(S-1)} \rangle = o_{ut} \delta_{vt} - o_{tv} \delta_{ut}$
15.	(S-1)(O1O1)	(S-1)(O1O1)	$\langle \Phi_u^{(S-1)(S-1)} \hat{O} \Phi_t^{(S-1)(S-1)} \rangle = 0$

TABLE V. Matrix elements $\langle \Phi_K^{S'S'} | \hat{O} | \Phi_L^{S'S'} \rangle$, where bra and ket are both $S' = S$

No.	Bra	Ket	Matrix element
16.	GS	GS	$\langle \Phi_0^{SS} \hat{O} \Phi_0^{SS} \rangle = 0$
17.	GS	S(CV0)	$\langle \Phi_0^{SS} \hat{O} \Phi_{bj,1}^{SS} \rangle = \sqrt{2} o_{bj}$
18.	GS	S(CO0)	$\langle \Phi_0^{SS} \hat{O} \Phi_{tj}^{SS} \rangle = o_{jt}$
19.	GS	S(OV0)	$\langle \Phi_0^{SS} \hat{O} \Phi_{bt}^{SS} \rangle = o_{tb}$
20.	GS	S(CV1)	$\langle \Phi_0^{SS} \hat{O} \Phi_{bj,2}^{SS} \rangle = 0$
21.	S(CV0)	S(CV0)	$\langle \Phi_{ai,1}^{SS} \hat{O} \Phi_{bj,1}^{SS} \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab}$
22.	S(CV0)	S(CO0)	$\langle \Phi_{ai,1}^{SS} \hat{O} \Phi_{tj}^{SS} \rangle = \frac{1}{\sqrt{2}} o_{at} \delta_{ij}$
23.	S(CV0)	S(OV0)	$\langle \Phi_{ai,1}^{SS} \hat{O} \Phi_{bt}^{SS} \rangle = -\frac{1}{\sqrt{2}} o_{ti} \delta_{ab}$
24.	S(CV0)	S(CV1)	$\langle \Phi_{ai,1}^{SS} \hat{O} \Phi_{bj,2}^{SS} \rangle = 0$
25.	S(CO0)	S(CO0)	$\langle \Phi_{ui}^{SS} \hat{O} \Phi_{vj}^{SS} \rangle = o_{ut} \delta_{ij} - o_{ji} \delta_{ut}$
26.	S(CO0)	S(OV0)	$\langle \Phi_{ui}^{SS} \hat{O} \Phi_{bv}^{SS} \rangle = 0$
27.	S(CO0)	S(CV1)	$\langle \Phi_{ui}^{SS} \hat{O} \Phi_{bj,2}^{SS} \rangle = \sqrt{\frac{1+S}{2S}} o_{ub} \delta_{ij}$
28.	S(OV0)	S(OV0)	$\langle \Phi_{au}^{SS} \hat{O} \Phi_{bt}^{SS} \rangle = o_{ab} \delta_{ut} - o_{tu} \delta_{ab}$
29.	S(OV0)	S(CV1)	$\langle \Phi_{au}^{SS} \hat{O} \Phi_{bj,2}^{SS} \rangle = \sqrt{\frac{1+S}{2S}} o_{ju} \delta_{ab}$
30.	S(CV1)	S(CV1)	$\langle \Phi_{ai,2}^{SS} \hat{O} \Phi_{bj,2}^{SS} \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab}$

TABLE VI. Matrix elements $\langle \Phi_K^{S'S'} | \hat{O} | \Phi_L^{S'S'} \rangle$, where bra and ket are both $S' = S + 1$

No.	Bra	Ket	Matrix element
31.	(S+1)(CV1)	(S+1)(CV1)	$\langle \Phi_{ai}^{(S+1)(S+1)} \hat{O} \Phi_{bj}^{(S+1)(S+1)} \rangle = o_{ab} \delta_{ij} - o_{ji} \delta_{ab}$

APPENDIX: TABLE OF THOSE EXCITED STATES

TABLE VII. Excited states with final total spin is $S' = S - 1, S, S + 1$, from a high-spin reference. The last state will be $|\Phi_u^{(S-1)(S-1)}\rangle = \sum_v |\Phi_v^v\rangle V_{vu}$ after orthonormalized.

S'	Type	Excited states	Expanded form	Ref.
$S + 1$	CV(1)	$[\hat{T}_{\text{CV}}^\dagger(1) \times S\rangle]_{S+1}^{S+1}$	$ \Phi_{ai}^{(S+1)(S+1)}\rangle = - \Phi_i^a\rangle$	Eq. (22)
S	CV(0)	$[\hat{S}_{\text{CV}}^\dagger(0) \times S\rangle]_S^S$	$ \Phi_{ai,1}^{SS}\rangle = \frac{1}{\sqrt{2}}(\Phi_i^a\rangle + \Phi_i^{\bar{a}}\rangle)$	Eq. (23)
	CO(0)	$\sqrt{2}[\hat{S}_{\text{CO}}^\dagger(0) \times S\rangle]_S^S$	$ \Phi_{ui}^{SS}\rangle = \Phi_i^{\bar{u}}\rangle$	Eq. (25)
	OV(0)	$\sqrt{2}[\hat{S}_{\text{OV}}^\dagger(0) \times S\rangle]_S^S$	$ \Phi_{au}^{SS}\rangle = \Phi_u^a\rangle$	Eq. (27)
	CV(1)	$[\hat{T}_{\text{CV}}^\dagger(1) \times S\rangle]_S^S$	$ \Phi_{ai,2}^{SS}\rangle = -\sqrt{\frac{S}{2(S+1)}}(\Phi_i^a\rangle - \Phi_i^{\bar{a}}\rangle) + \frac{1}{S} \sum_x \Phi_{ix}^{a\bar{x}}\rangle$	Eq. (29)
$S - 1$	CV(1)	$[\hat{T}_{\text{CV}}^\dagger(1) \times S\rangle]_{S-1}^{S-1}$	$ \Phi_{ai}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S+1}} \Phi_i^{\bar{a}}\rangle - \frac{1}{2S}\sqrt{\frac{2S-1}{2S+1}}\sum_x(\Phi_{ix}^{a\bar{x}}\rangle - \Phi_{ix}^{\bar{a}\bar{x}}\rangle) - \frac{1}{2S\sqrt{(2S+1)(2S-1)}}\sum_{xy} \Phi_{ixy}^{a\bar{x}y}\rangle$	Eq. (38)
	CO(1)	$\sqrt{\frac{2S}{2S+1}}[\hat{T}_{\text{CO}}^\dagger(1) \times S\rangle]_{S-1}^{S-1}$	$ \Phi_{ui}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}} \Phi_i^{\bar{u}}\rangle + \frac{1}{\sqrt{2S(2S-1)}}\sum_{x \neq u} \Phi_{ix}^{\bar{u}\bar{x}}\rangle$	Eq. (52)
	OV(1)	$\sqrt{\frac{2S}{2S+1}}[\hat{T}_{\text{OV}}^\dagger(1) \times S\rangle]_{S-1}^{S-1}$	$ \Phi_{au}^{(S-1)(S-1)}\rangle = \sqrt{\frac{2S-1}{2S}} \Phi_u^{\bar{a}}\rangle - \frac{1}{\sqrt{2S(2S-1)}}\sum_{x \neq u} \Phi_{ux}^{a\bar{x}}\rangle$	Eq. (59)
	$\text{O}_1\text{O}_2(1)$	$\sqrt{\frac{2S-1}{2S+1}}[\hat{T}_{\text{O}_1\text{O}_2}^\dagger(1) \times S\rangle]_{S-1}^{S-1}$	$ \Phi_{uv}^{(S-1)(S-1)}\rangle = \Phi_v^{\bar{u}}\rangle$	Eq. (64)
	$\text{O}_1\text{O}_1(1)$	$\sqrt{\frac{2S-1}{2S+1}}[\hat{T}_{\text{O}_1\text{O}_1}^\dagger(1) \times S\rangle]_{S-1}^{S-1}$	$ \Phi_u^{(S-1)(S-1)}\rangle = \Phi_u^{\bar{u}}\rangle - \frac{1}{2S}\sum_x \Phi_x^{\bar{x}}\rangle$	Eq. (87)

REFERENCES